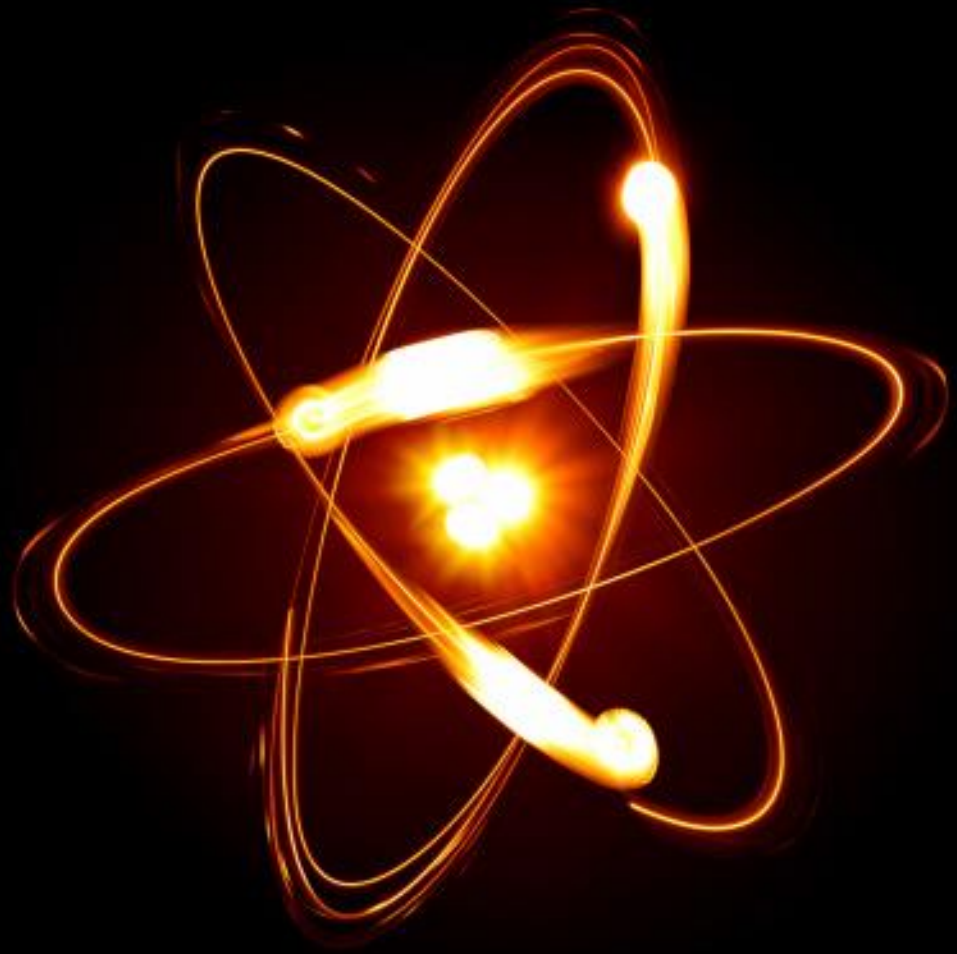
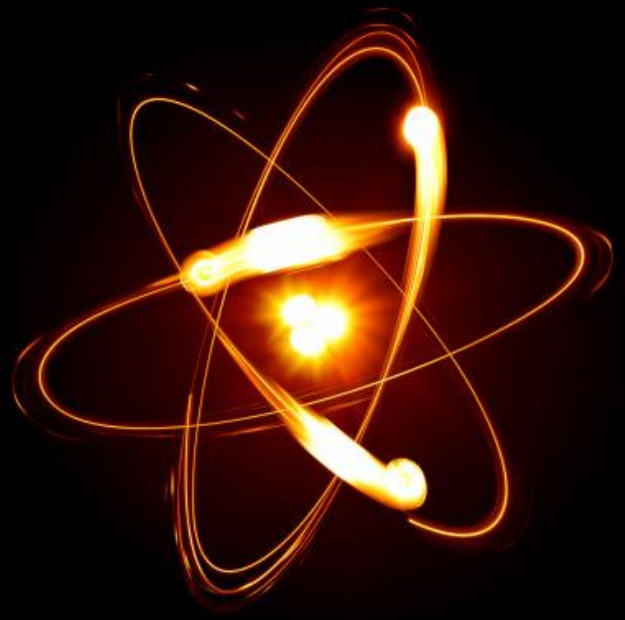
Chapter 8.

One-Electron Atoms



Beatriz González del Río

Quantum Physics



Chapter 8.

One-Electron Atoms

- Schrödinger´s equation in three dimensions
- 2-D and 3-D Square Well
- 3-D Harmonic oscillator
- Curvilinear coordinates
- Central Potentials
- The two bodies' problem
- Hydrogen's atom: eigenfunctions, eigenvalues, degeneracy and probability density

Schrödinger's equation in three dimensions

The interesting (and real) problems are in three dimensions

Schrödinger's equation in three dimensions:

$$i\hbar \frac{\partial \Psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Psi(\vec{r},t) + V(\vec{r},t)\Psi(\vec{r},t)$$

Normalization

$$\int_{all\ space} |\Psi(\vec{r},t)|^2\, d^3\vec{r} = 1$$

Schrödinger's equation in three dimensions

Simplify
If the potential is time-independent, i.e. $V(\vec{r},t) \equiv V(\vec{r})$, then the solution can be decomposed as:

$$\Psi(\vec{r},t) = \psi(\vec{r})\phi(t)$$

going back to Schrödinger's equation and operating, we finally obtain

$$-i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = \underbrace{E}_{\text{constant}} = \left[-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r})\right]\frac{1}{\psi(\vec{r})}$$

The solution for the time-dependent part is:

$$\phi(t) = e^{-\frac{iEt}{\hbar}}$$

Schrödinger's equation in three dimensions

and $\psi(\vec{r})$ is the solution for:

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r})$$

which is the time-independent Schrödinger's equation

It must be solved along with the condition

$$\int_{all\ space} |\psi(\vec{r})|^2\, d^3\vec{r} = 1$$

Separation of variables

- Separation of variables in cartesian coordinates:

If $V(\vec{r}) = V_1(x) + V_2(y) + V_3(z)$, then the solution $\psi(\vec{r})$ can be factorized, i.e.

$$\psi(\vec{r}) = X(x)Y(y)Z(z)$$

which leads to:

$$\left[-\frac{\hbar^2}{2m}\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} + V_1(x)\right] + \left[-\frac{\hbar^2}{2m}\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} + V_2(y)\right] + \left[-\frac{\hbar^2}{2m}\frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} + V_3(z)\right] = E$$

and this leads to the following three equations:

Separation of variables

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2} + V_1(x)X(x) = E_1X(x)$$

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2} + V_2(y)Y(y) = E_2Y(y)$$

$$-\frac{\hbar^2}{2m}\frac{d^2Z(z)}{dz^2} + V_3(z)Z(z) = E_3Z(z)$$

with $E = E_1 + E_2 + E_3$

Separation of variables

Examples:

- 2-dimensional infinite square well
- 3-dimensional infinite square well
- 3-D harmonic oscillator

1-D infinite square well

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & otherwise \end{cases}$$

$$\phi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$$

Remember !!! **Orthonormality condition**

$$\int_0^a dx \ \phi_n^*(x)\phi_{n'} = \delta_{n,n'}$$

2-D square well

In 2-D

$$V(x,y) = \begin{cases} 0 & 0 < x < a \quad and \quad 0 < y < b \\ \infty & otherwise \end{cases}$$

$$V(x,y) = V_1(x) + V_2(y)$$

$$V_1(x) = \begin{cases} 0 & 0 < x < a \\ \infty & otherwise \end{cases} \qquad V_2(y) = \begin{cases} 0 & 0 < y < b \\ \infty & otherwise \end{cases}$$

$$\phi_{n_x,n_y}(x,y) = \sqrt{\frac{2}{a}}\sin\left(\frac{n_x\pi x}{a}\right)\sqrt{\frac{2}{b}}\sin\left(\frac{n_y\pi y}{b}\right) \qquad E_{n_x,n_y} = E_{n_x} + E_{n_y} = \frac{\hbar^2\pi^2 n_x^2}{2ma^2} + \frac{\hbar^2\pi^2 n_y^2}{2mb^2} \qquad (n_x, n_y = 1,2,3,\dots)$$

Remember !!! **Orthonormality condition**

$$\int_0^a dx \int_0^b dy \quad \phi^*_{n_x,n_y}(x,y)\phi_{n'_x,n'_y}(x,y) = \delta_{n_x,n'_x}\delta_{n_y,n'_y}$$

3-D square well

In 3-D

$$\phi_{n_x,n_y,n_z}(x,y,z) = \sqrt{\frac{2}{a}}\sin\left(\frac{n_x\pi x}{a}\right)\sqrt{\frac{2}{b}}\sin\left(\frac{n_y\pi y}{b}\right)\sqrt{\frac{2}{c}}\sin\left(\frac{n_z\pi z}{c}\right)$$

$$E_{n_x,n_y,n_z} = E_{n_x} + E_{n_y} + E_{n_z} = \frac{\hbar^2\pi^2 n_x^2}{2ma^2} + \frac{\hbar^2\pi^2 n_y^2}{2mb^2} + \frac{\hbar^2\pi^2 n_z^2}{2mc^2} \qquad (n_x, n_y, n_z = 1,2,3,\dots)$$

Remember !!! **Orthonormality condition**

$$\int_0^a dx \int_0^b dy \int_0^c dz \quad \phi^*_{n_x,n_y,n_z}(x,y,z)\phi_{n'_x,n'_y,n'_z}(x,y,z) = \delta_{n_x,n'_x}\delta_{n_y,n'_y}\delta_{n_z,n'_z}$$

2-D and 3-D square well: degeneration

If we consider the case $a = b = c$,

$$\phi_{n_x,n_y,n_z}(x,y,z) = \sqrt{\frac{2}{a}}\sin\left(\frac{n_x\pi x}{a}\right)\sqrt{\frac{2}{a}}\sin\left(\frac{n_y\pi y}{a}\right)\sqrt{\frac{2}{a}}\sin\left(\frac{n_z\pi z}{a}\right)$$

$$E_{n_x,n_y,n_z} = E_{n_x} + E_{n_y} + E_{n_z} = \frac{\hbar^2\pi^2 n_x^2}{2ma^2} + \frac{\hbar^2\pi^2 n_y^2}{2ma^2} + \frac{\hbar^2\pi^2 n_z^2}{2ma^2} = \frac{\hbar^2\pi^2(n_x^2 + n_y^2 + n_z^2)}{2ma^2}$$

Now we have degeneration for the energy levels (it may happen that)

$$E_{n_x,n_y,n_z} = E_{n_y,n_z,n_x} = E_{n_z,n_x,n_y} = \cdots$$

$$\phi_{n_x,n_y,n_z}(x,y,z) \neq \phi_{n_y,n_z,n_x}(x,y,z) \neq \phi_{n_z,n_x,n_y}(x,y,z) \ldots$$

and

$$\int_0^a dx \int_0^b dy \int_0^c dz \quad \phi^*_{n_x,n_y,n_z}(x,y,z)\phi_{n_y,n_z,n_x}(x,y,z) = 0$$

Particle in a box

Degeneration: Different combinations of the quantum numbers that give the same value for the energy.

There are different eigenfunctions with the same energy value

Degeneration of the energy levels in a cubic box of side a

Energy Levels and Degeneracies in a Cubical Box ($E_1 = \pi^2\hbar^2/2ma^2$)

Energy	Combinations of n_1, n_2, n_3	Degeneracy, g
$3E_1$	(1, 1, 1)	1
$6E_1$	(2, 1, 1)(1, 2, 1)(1, 1, 2)	3
$9E_1$	(2, 2, 1)(2, 1, 2)(1, 2, 2)	3
$11E_1$	(3, 1, 1)(1, 3, 1)(1, 1, 3)	3
$12E_1$	(2, 2, 2)	1
$14E_1$	(1, 2, 3)(3, 2, 1)(2, 3, 1) (1, 3, 2)(2, 1, 3)(3, 1, 2)	6

2-D and 3-D square well: Summary

$$E_{n_x,n_y,n_z} = E_{n_x} + E_{n_y} + E_{n_z} = \frac{\hbar^2\pi^2 n_x^2}{2ma^2} + \frac{\hbar^2\pi^2 n_y^2}{2mb^2} + \frac{\hbar^2\pi^2 n_z^2}{2mc^2}$$

$$\phi_{n_x,n_y,n_z}(x,y,z) = \sqrt{\frac{2}{a}}\sin\left(\frac{n_x\pi x}{a}\right)\sqrt{\frac{2}{b}}\sin\left(\frac{n_y\pi y}{b}\right)\sqrt{\frac{2}{c}}\sin\left(\frac{n_z\pi z}{c}\right)$$

$$\int_0^a dx \int_0^b dy \int_0^c dz \quad \phi^*_{n_x,n_y,n_z}(x,y,z)\phi_{n'_x,n'_y,n'_z}(x,y,z) = \delta_{n_x,n'_x}\delta_{n_y,n'_y}\delta_{n_z,n'_z}$$

Particle in a box

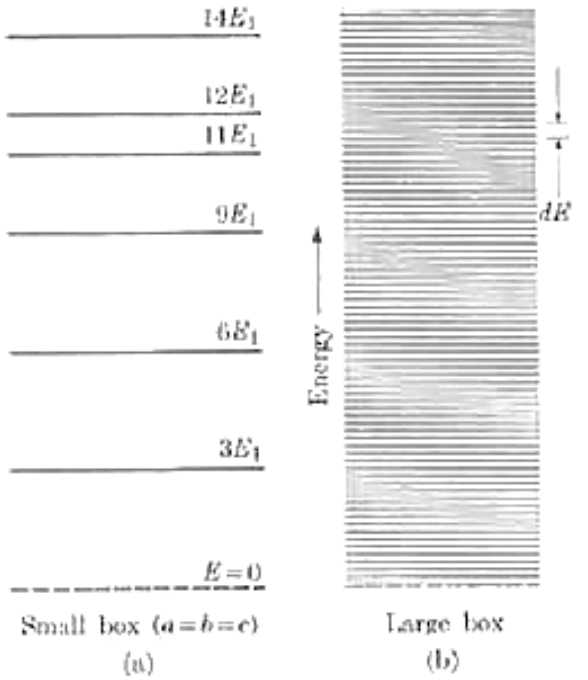
(See M Alonso and E Finn, *Fisica, Volumen III. Fundamentos cuanticos y estadisticos*)

Number of energy levels in a small energy range *dE* for a particle in a very large potential box

For a small box, the energy levels are widely separated

For a very large box, the energy levels are so close that practically are a continuous spectrum (i.e. molecules of a gas in a container, electrons in a metal)

How many energy levels are there within an energy range, *dE* , centered around a given value E ?



Energy levels for (a) a small potential box, (b) a large potential box.

Particle in a box

Number of energy levels in a small energy range *dE* for a particle in a very large potential box

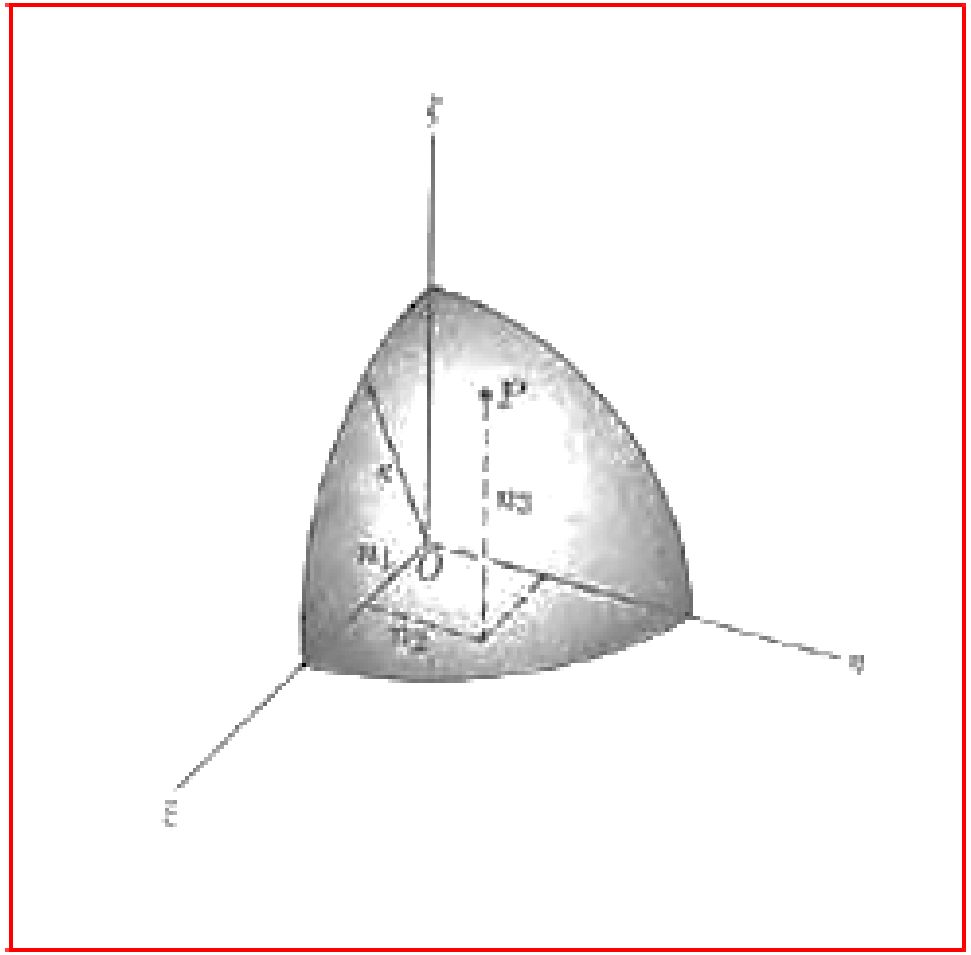
Introduce coordinates ξ, η, ζ

Each point $\xi = n_1$, $\eta = n_2$, and $\zeta = n_3$ represents an energy level

$$\kappa^2 = \xi^2 + \eta^2 + \zeta^2$$

The number of points on the surface of a sphere of radius κ give the number of different states with energy

$$E = \frac{\pi^2 \hbar^2}{2ma^2} \kappa^2$$



Particle in a box

Number of states, $N(E)$, with energy between cero and E, equals the volume of a (octant) sphere

$$N(E) = \frac{1}{8}\left(\frac{4}{3}\pi\kappa^3\right) = \frac{8\pi V}{3h^3}\left(2m^3\right)^{1/2}E^{3/2} \qquad V = a^3$$

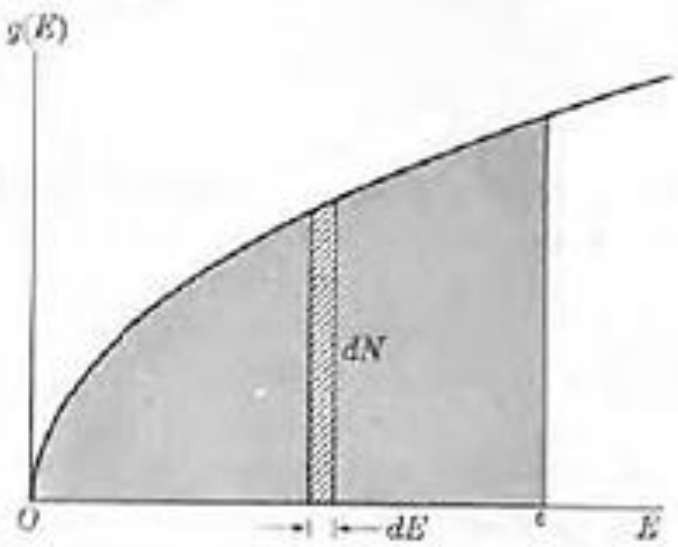
The number of states with energy in the range $[E, E + dE]$ is

$$dN(E) = \frac{4\pi V\left(2m^3\right)^{1/2}}{h^3}E^{1/2}dE$$

Writing $dN(E) = g(E)dE$

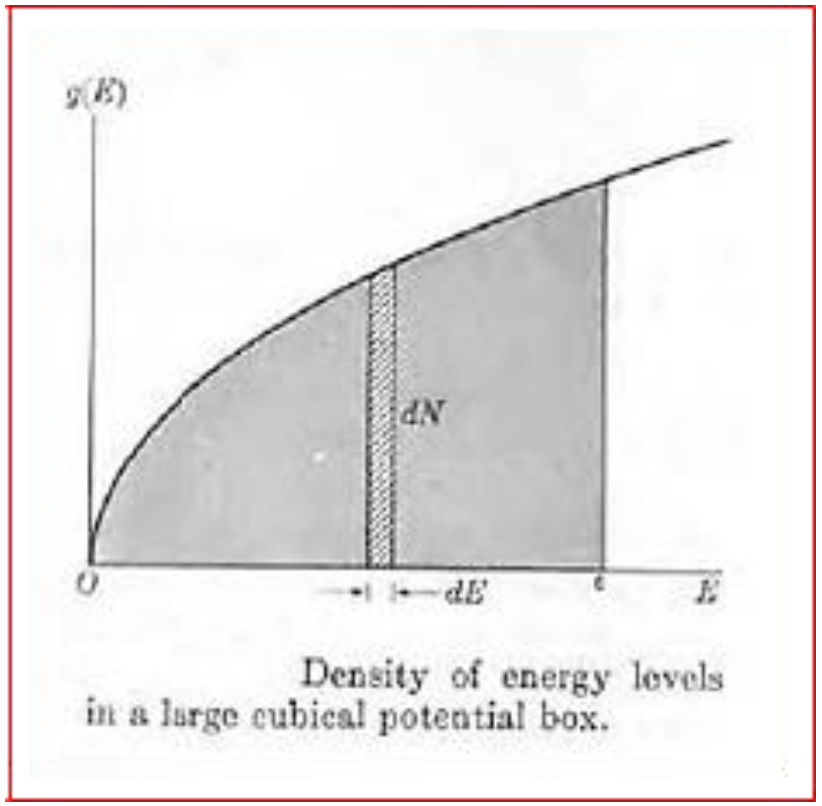
$$g(E) = \frac{dN}{dE} = \frac{4\pi V\left(2m^3\right)^{1/2}}{h^3}E^{1/2}$$

is the number of states per unit energy interval at the energy E.
(density of states)



Density of energy levels in a large cubical potential box.

Particle in a box



Density of energy levels in a large cubical potential box.

The area of a strip located at E and width dE gives the number $dN = g(E)dE$ of states in such an energy state, $[E, E + dE]$.

The area under the curve from E=0 to $E = c$ gives the total number of states with energies in $[0, c]$.

Number of states with momentum in the range $[p, p + dp]$, that is, $g(p)dp$

$$g(p) = g\big(E(p)\big)\left|\frac{dE}{dp}\right| = \frac{4\pi V}{h^3}p^2$$

Remember that $E = p^2/2m$

The 1-D Harmonic Oscillator

$$V(x) = \frac{1}{2}kx^2$$

Solution:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad \omega = \sqrt{k/m}$$

$$\phi_n(x) = A_n e^{-\frac{\beta^2 x^2}{2}} H_n(\beta x) \qquad \beta^2 = \frac{m\omega}{\hbar}$$

$H_n(u)$ are the Hermite's polynomials $\qquad n = 0,1,2,\ldots$

$$H_0(u) = 1 \qquad H_2(u) = 2 - 4u^2$$

$$H_1(u) = 2u \qquad H_3(u) = 12u - 8u^3$$

The 3-D Harmonic Oscillator

Consider a 3-D harmonic oscillator (non-isotropic):

$$V(x,y,z) = \frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2 + \frac{1}{2}k_z z^2$$

Solution:

$$E_{n_x,n_y,n_z} = E_{n_x}^x + E_{n_y}^y + E_{n_z}^z = \left(n_x + \frac{1}{2}\right)\hbar\omega_x + \left(n_y + \frac{1}{2}\right)\hbar\omega_y + \left(n_z + \frac{1}{2}\right)\hbar\omega_z \qquad n_x, n_y, n_z = 0,1,2,\ldots$$

$$\phi_{n_x,n_y,n_z}(x,y,z) = C_{n_x,n_y,n_z} e^{-\frac{\beta_x^2 x^2}{2}} H_{n_x}(\beta_x x) e^{-\frac{\beta_y^2 y^2}{2}} H_{n_y}(\beta_y y)\, e^{-\frac{\beta_z^2 z^2}{2}} H_z(\beta_z z) \qquad \beta_{x,y,z}^2 = \frac{m\omega_{x,y,z}}{\hbar}$$

If $k_x = k_y = k_z$ (isotropic oscillator)

$$E_{n_x,n_y,n_z} = E_{n_x}^x + E_{n_y}^y + E_{n_z}^z = \left(n_x + n_y + n_z + \frac{3}{2}\right)\hbar\omega$$

$$\phi_{n_x,n_y,n_z}(x,y,z) = C_{n_x,n_y,n_z} e^{-\frac{\beta^2(x^2+y^2+z^2)}{2}} H_{n_x}(\beta x) H_{n_y}(\beta y) H_{n_z}(\beta z)$$

Curvilinear Coordinates. Summary

Let's consider a transformation of the cartesian coordinates x,y,z, to the u,v,w coordinates given by

$$x = f(u, v, w)$$

$$y = g(u, v, w)$$

$$z = h(u, v, w)$$

How is the expression of

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Curvilinear Coordinates. Summary

$$\nabla^2 = \frac{1}{q_u q_v q_w}\left(\frac{\partial}{\partial u}\left(\frac{q_v q_w}{q_u}\frac{\partial}{\partial u}\right) + \frac{\partial}{\partial v}\left(\frac{q_u q_w}{q_v}\frac{\partial}{\partial v}\right) + \frac{\partial}{\partial w}\left(\frac{q_u q_v}{q_u}\frac{\partial}{\partial u}\right)\right)$$

where

$$q_u^2 = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2$$

$$q_v^2 = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2$$

$$q_w^2 = \left(\frac{\partial x}{\partial w}\right)^2 + \left(\frac{\partial y}{\partial w}\right)^2 + \left(\frac{\partial z}{\partial w}\right)^2$$

Curvilinear Coordinates. Summary

The differential element of volume is:

$$d\tau = dx\, dy\, dz = q_u\, q_v\, q_w du\, dv\, dw$$

Mathematically, it has been proved that the three-dimensional Schrödinger's equation can be separated only in a limited set of coordinate systems as long as,

$$\mathrm{V} = q_u V_u(u) + q_v V_v(v) + q_w V_w(w)$$

Curvilinear Coordinates. Examples

Cylindrical Coordinates

$$x = \rho \cos(\varphi)$$

$$y = \rho \sin(\varphi)$$

$$z = z$$

$$q_p = 1, \qquad q_z = 1, \qquad q_\varphi = \rho$$

$$d\tau = \rho \, d\rho \, dz \, d\varphi$$

$$\nabla^2 = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2}$$

Curvilinear Coordinates. Examples

Spherical Coordinates

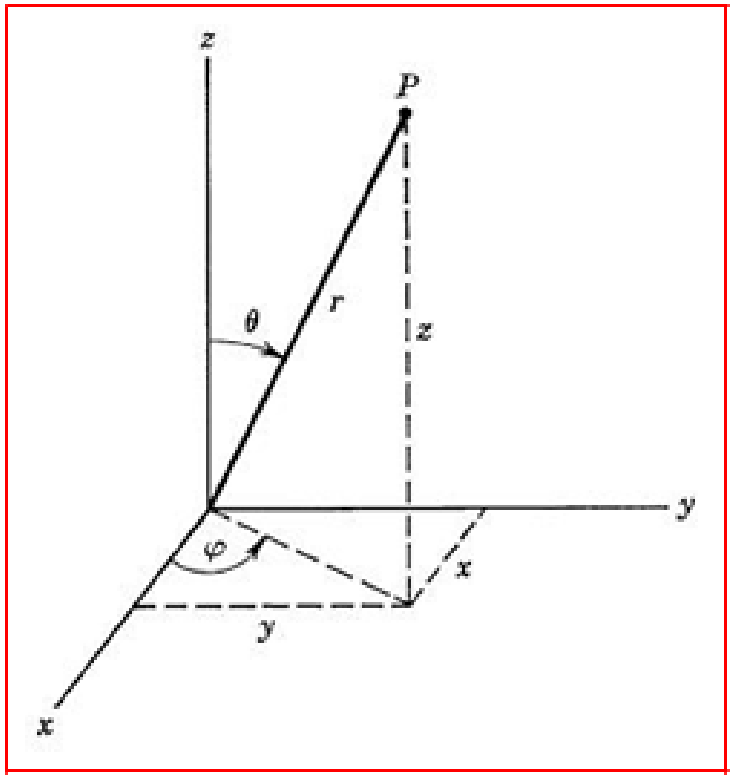
$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r \cos\theta$$

$$q_r = 1, \qquad q_\theta = r, \qquad q_\phi = r \sin\theta$$

$$d\tau = r^2 \sin\theta \; dr \, d\theta \, d\phi$$



$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial \phi^2}$$

Spherical Coordinates

There are some problems for which, by symmetry reasons, it is more convenient to use spherical coordinates.

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

The equation (for particle of mass μ)

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r})$$

can be written as:

$$-\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] + V(r,\theta,\phi)\psi = E\psi$$

and the normalization condition becomes:

$$\int_0^\infty\int_0^\pi\int_0^{2\pi} r^2\sin\theta\ dr\,d\theta\,d\phi\ |\psi(r,\theta,\phi)|^2 = 1$$

Central Potentials

$$V \equiv V(r)$$

central potential !!!

Again, we factorize the solution: $\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$

Substitute in SE divide by ψ and multiply by r^2

$$\underbrace{\left[-\frac{\hbar^2}{2\mu}\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+r^2V-r^2E\right]}_{=-\lambda}+\underbrace{\left[-\frac{\hbar^2}{2\mu}\frac{1}{Y}\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)-\frac{\hbar^2}{2\mu}\frac{1}{Y}\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]}_{=\lambda}=0$$

which can be rewritten as

Central Potentials

$$\underbrace{-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left(V(r)+\frac{\lambda}{r^2}\right)R=ER}$$

Radial equation

$$\underbrace{-\frac{\hbar^2}{2\mu}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]=\lambda Y}$$

Angular equation – (universal)

This solution applies to any spherically symmetric potential

Central Potentials

Separation process $Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$

$$\underbrace{\left[-\frac{\hbar^2}{2\mu}\frac{\sin\theta}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) - \lambda\sin^2\theta\right]}_{=-\nu} + \underbrace{\left[-\frac{\hbar^2}{2\mu}\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}\right]}_{=\nu} = 0$$

$$-\frac{\hbar^2}{2\mu}\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) - \lambda\sin^2\theta\Theta + \nu\Theta = 0 \qquad -\frac{\hbar^2}{2\mu}\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = \nu$$

which has the solution

$$\Phi(\phi) = Ae^{i\left(2\mu\nu/\hbar^2\right)^{1/2}\phi} + Be^{-i\left(2\mu\nu/\hbar^2\right)^{1/2}\phi} \qquad \nu \neq 0$$

$$\Phi(\phi) = A + B\phi \qquad \nu = 0$$

Central Potentials

$\Phi(\phi)$ and its first derivative must be continuous

$$\Phi(2\pi) = \Phi(0) \qquad \left.\frac{d}{d\phi}\Phi\right|_{2\pi} = \left.\frac{d}{d\phi}\Phi\right|_{0}$$

which finally leads to the condition

$$\left(2\mu v/\hbar^2\right)^{1/2} = m$$

m integer positive, cero or negative.

Finally, the (normalized) physical solutions are written as

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \qquad m = 0, \pm 1, \pm 2, \pm 3, \ldots$$

Obviously $\int_0^{2\pi} |\Phi(\phi)|^2 d\phi = 1$

Central Potentials

Now, we solve the other differential eqn.

$$-\frac{\hbar^2}{2\mu}\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)-\lambda\sin^2\theta\Theta+\nu\Theta=0$$

Returning to the eqn. for Θ and substituting the value for ν

$$\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda'\sin^2\theta-m^2\right)\Theta=0 \qquad \text{with } \lambda'=2\mu\lambda/\hbar^2$$

Changing variables $\cos\theta\equiv v$ and $\Theta(\theta)\equiv P(v)$ and taking into account that

$$\frac{d}{d\theta}=-\sin\theta\frac{d}{dv}=-\left(1-v^2\right)^{1/2}\frac{d}{dv}$$

leading to the new differential eqn.

$$\frac{d}{dv}\left[\left(1-v^2\right)\frac{dP}{dv}\right]+\left[\lambda'-\frac{m^2}{1-v^2}\right]P(v)=0 \qquad (**)$$

Case: $m = 0$

$$\frac{d}{dv}\left[\left(1-v^2\right)\frac{dP}{dv}\right]+\lambda'P(v)=0 \qquad \Rightarrow \qquad P(v)=\sum_{p=0}^{\infty}a_p v^p$$

Central Potentials

We obtain

$$\sum_{p=0}^{\infty}\left[a_{p+2}(p+2)(p+1)-a_p(p(p+1)-\lambda')\right]v^p=0 \qquad \frac{a_{p+2}}{a_p}=\frac{p(p+1)-\lambda'}{(p+1)(p+2)}$$

This ratio goes to 1 when $p \to \infty$

In fact

$$P(v)=a_0\left(1+\frac{a_2}{a_0}v^2+\frac{a_4}{a_2}\frac{a_2}{a_0}v^4+\cdots\right)+a_1\left(v+\frac{a_3}{a_1}v^3+\frac{a_5}{a_3}\frac{a_3}{a_1}v^5+\cdots\right)$$

Both series behave (when $p \to \infty$) as the expansion of $1/(1-u^2)$ which diverges when $u \to \pm 1$. Therefore, they MUST be terminated at some finite value, i.e. $p=l$; this leads to the condition

$$p(p+1)-\lambda'=0 \qquad \Longrightarrow \qquad \lambda'=l(l+1)$$

Central Potentials

$\lambda' = l(l+1) \qquad l \geq 0$

with $l = 0,2,4,6,\ldots$ and $a_1 = 0$

with $l = 1,3,5,7,\ldots$ and $a_0 = 0$

Now, $P(v) \equiv P_l(v)$ is a polynomial of degree l with only odd (or even) powers. (Legendre polynomials)

$$P_0(v) = 1 \qquad P_1(v) = v \qquad P_2(v) = \frac{1}{2}(3v^2 - 1) \qquad P_3(v) = \frac{1}{2}(5v^3 - 3v)$$

$$P_l(v) = \frac{(-1)^l}{2^l\, l!}\frac{d^l}{dv^l}\left[(1-v^2)^l\right]$$

Case: $m \neq 0$

Eqn $(**)$ depends on m^2, so the solution does not depend on the sign of m. The solution will be characterized by l and $|m|$ and is written as $P_l^{|m|}(v)$

$$P_l^{|m|}(v) = (1-v^2)^{|m|/2}\frac{d^{|m|}P_l}{dv^{|m|}}$$

This eqn. imposes some condition on the allowed values for m. P_l is a polynomial of degree l, so its derivative will be zero if $|m| > l$. In this case, the wavefunction would be zero over all space, which is physically unrealistic. Therefore

$-l \leq m \leq l$ or $|m| \leq l$

Central Potentials

Combining the solutions for Φ and Θ, such that $Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$

$$Y_l^m(\theta,\phi) = \left[\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}\right]^{1/2} P_l^m(\cos\theta)e^{im\phi} \qquad m \geq 0$$

with

$$Y_l^{-m}(\theta,\phi) = (-1)^m[Y_l^m(\theta,\phi)]^* \qquad \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \ \sin\theta \ |Y_l^m(\theta,\phi)|^2 = 1$$

$$Y_{00} = \frac{1}{(4\pi)^{1/2}} \qquad Y_{10} = \left(\frac{3}{4\pi}\right)^{1/2}\cos\theta \qquad Y_{1\pm1} = \mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta\, e^{\pm i\phi}$$

$$Y_{20} = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1) \qquad Y_{2\pm1} = \mp\left(\frac{15}{8\pi}\right)^{1/2}\cos\theta\sin\theta\, e^{\pm i\phi} \qquad Y_{2\pm2} = \left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta\, e^{\pm 2i\phi}$$

Central Potentials

Another way

$$P_l^m = (-1)^m \left(1 - v^2\right)^{\frac{m}{2}} \frac{d^m}{dv^m} P_l(v) \qquad m \geq 0$$

and

$$P_l^m = (-1)^{|m|} \frac{(l - |m|)!}{(l + |m|)!} P_l^{|m|}(v) \qquad m < 0$$

$$\int_0^\pi P_l^m(\cos\theta) P_{l'}^m(\cos\theta) \sin\theta \, d\theta = \int_{-1}^1 P_l^m(v) P_{l'}^m(v) dv = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \delta_{ll'}$$

and then the spherical harmonics: $Y_l^m(\theta, \phi)$,

$$Y_l^m(\theta, \phi) = \left[\frac{(2l+1)}{4\pi} \frac{(l-m)!}{(l+m)!}\right]^{\frac{1}{2}} P_l^m(\cos\theta) e^{im\phi}$$

Central Potentials

Some properties of the spherical harmonics:

$$[Y_l^m(\theta,\phi)]^* = (-1)^m Y_l^{-m}(\theta,\phi)$$

$$\sum_{m=-l}^{l} |Y_l^m(\theta,\phi)|^2 = \frac{2l+1}{4\pi} \qquad (\text{fixed } l, \quad \forall\, l \geq 0)$$

$$\int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta\, [Y_l^m(\theta,\phi)]^* Y_{l'}^{m'}(\theta,\phi) = \delta_{l,l'}\delta_{m,m'}$$

Moreover, any bounded function $f(\theta,\phi)$, can be expanded

$$f(\theta,\phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{m=l} c_{l,m} Y_l^m(\theta,\phi) \qquad c_{l,m} = \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta\, [Y_l^m(\theta,\phi)]^* f(\theta,\phi)$$

The spherical harmonics are an orthonormal basis for the functions $f(\theta,\phi)$

Central Potentials

Remember that $Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$

Angular dependence of the probability density functions ?

$$\psi^*(r,\theta,\phi)\,\psi(r,\theta,\phi) = R^*_{n,l}(r)\Theta^*_{l,m}(\theta)\Phi^*_m(\phi)\,R_{n,l}(r)\Theta_{l,m}(\theta)\Phi_m(\phi)$$

- $R^*_{n,l}(r)R_{n,l}(r)$ depends on the radial distance only
- $\Phi^*_m(\phi)\Phi_m(\phi) = \frac{1}{2\pi}$
- $\Theta^*_{l,m}(\theta)\Theta_{l,m}(\theta)$ carries the dependence on direction
- Plot of $\Theta^*_{l,m}(\theta)\Theta_{l,m}(\theta)$ in polar coordinates
- Distance from origin = $\Theta^*_{l,m}(\theta)\Theta_{l,m}(\theta)$

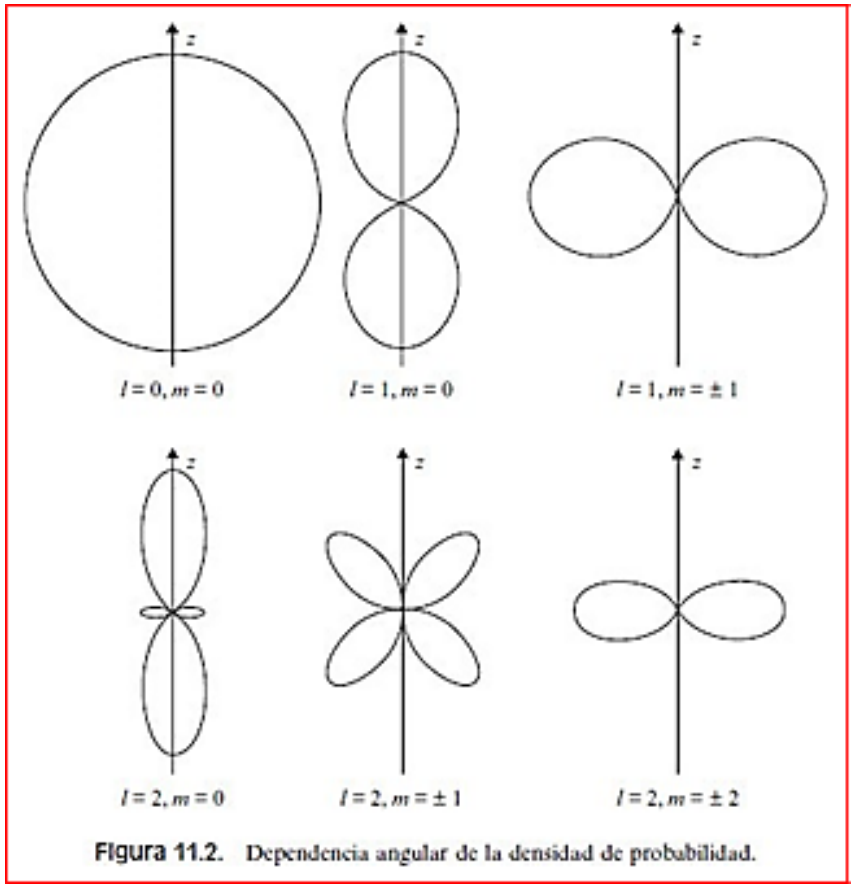


Figura 11.2. Dependencia angular de la densidad de probabilidad.

Central Potentials

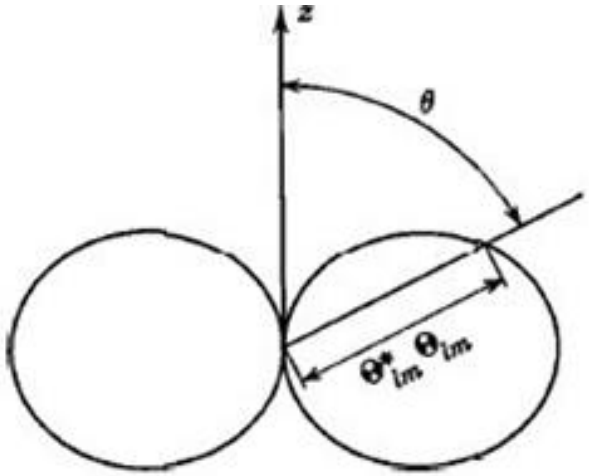
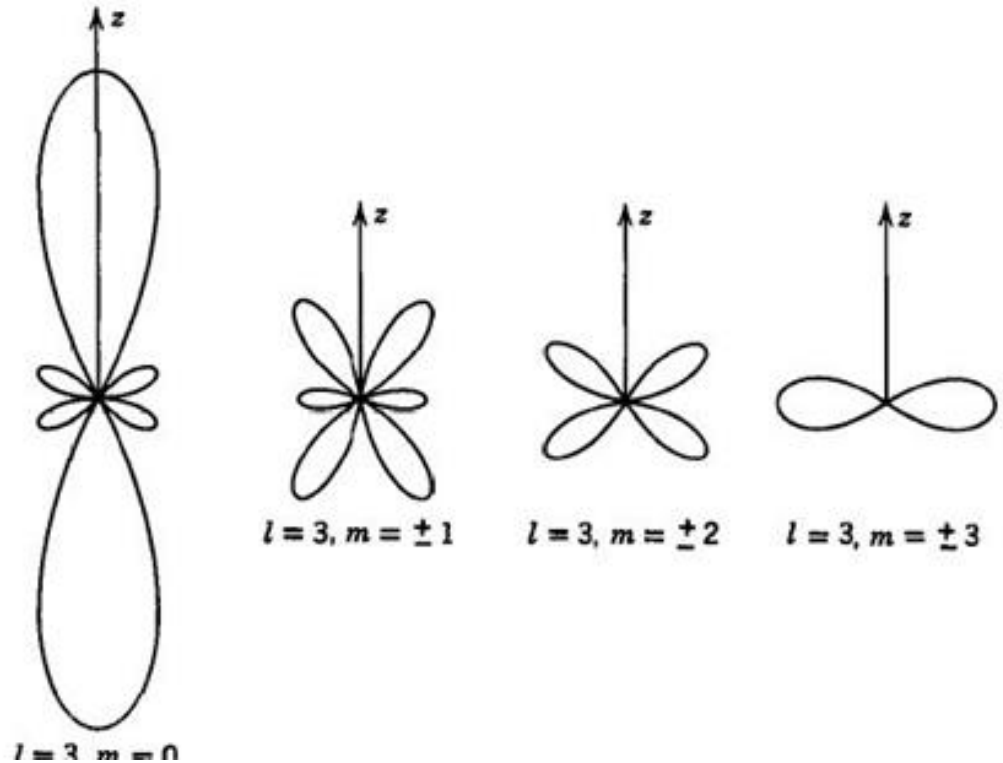


Figure 10–3. A polar diagram of the directionally dependent modulation factor of the probability density.



Central Potentials

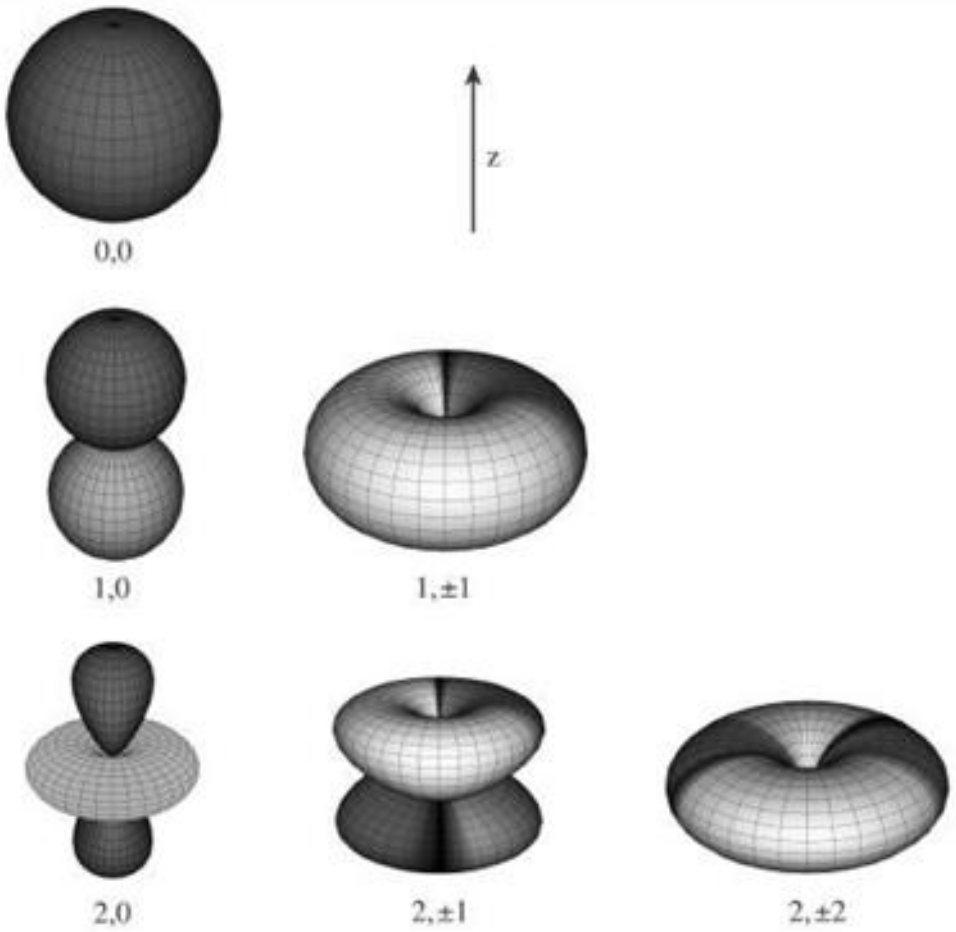
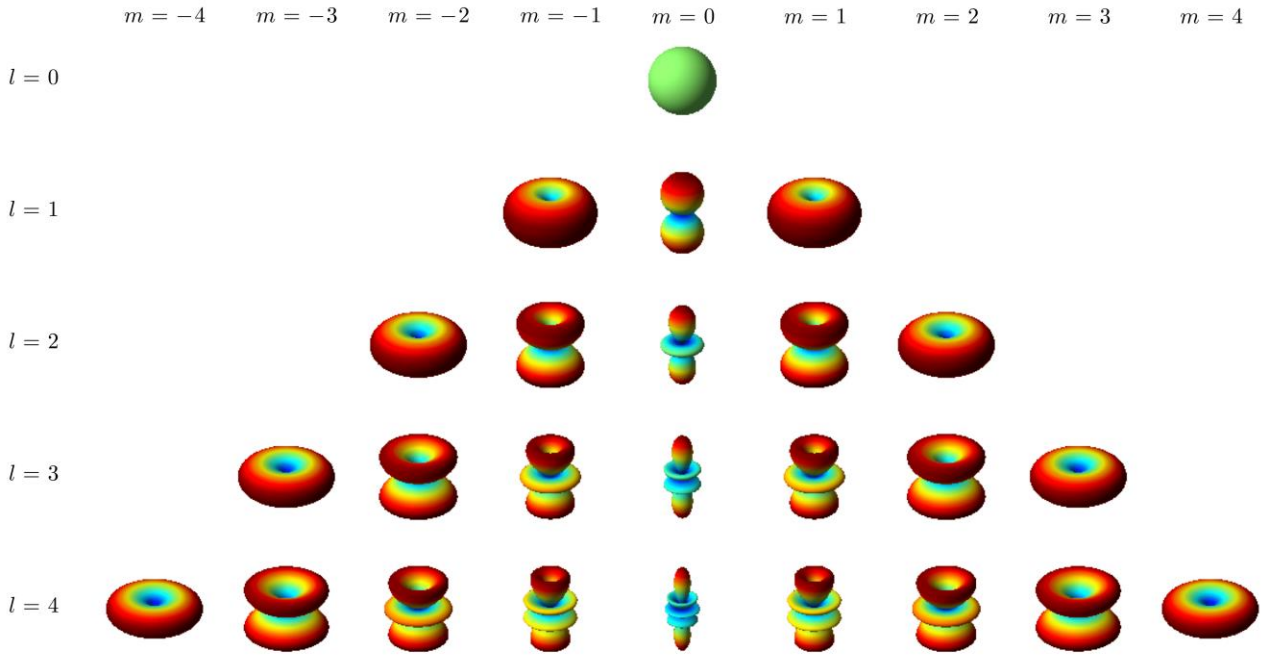


Figure 3.3. Representations of the shapes of the spherical harmonics with quantum numbers l, m, where $l \leqslant 2$ and the z axis is vertical. In the case of $m = 0$, the dark and light regions have opposite sign; when $m \neq 0$, the function is complex and its phase changes by $2m\pi$ during a complete circuit of the z axis.



Central Potentials

When solving the radial equation

$$-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right)+\left(V(r)+\frac{\lambda}{r^2}\right)R(r)=ER(r) \quad \Rightarrow \quad -\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right)+\left(V(r)+\frac{l(l+1)\hbar^2}{2\mu r^2}\right)R(r)=ER(r)$$

Let us write $\mathcal{X}(r)=rR(r)$

$$-\frac{\hbar^2}{2\mu}\frac{d^2\mathcal{X}(r)}{dr^2}+\left(V(r)+\frac{l(l+1)\hbar^2}{2\mu r^2}\right)\mathcal{X}(\mathrm{r})=E\mathcal{X}(r)$$

Notice that $|\mathcal{X}(r)|^2dr$ is the probability of finding the particle at a distance between *r* and *r* + *dr*

$$\left|R^2(r)\right|r^2dr\int_0^{2\pi}\int_0^{\pi}|Y(\theta,\phi)|^2\sin\theta\,d\theta d\phi=\left|\mathcal{X}^2(r)\right|dr$$

Notice that we obtain a different solution for each central potential

The two bodies' problem

(See Carlos Sanchez del Rio (coordinador), *Fisica Cuantica*) ,

System made of two (distinguishable) particles. Its wave function is

$$\psi(\vec{r}_1, \vec{r}_2, t) \equiv \psi(x_1, y_1, z_1, x_2, y_2, z_2, t)$$

Probability of finding, at time t, particle 1 inside $d\vec{r}_1$ centered around $\vec{r}_1$ and particle 2 inside $d\vec{r}_2$ centered around $\vec{r}_2$?

$$|\psi(\vec{r}_1, \vec{r}_2, t)|^2 \, d\vec{r}_1 \, d\vec{r}_2$$

How to evaluate $\psi(\vec{r}_1, \vec{r}_2, t)$? Solve its Schrödinger's eqn.

$$\left[-\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{\hbar^2}{2m_2} \nabla_2^2 + V(\vec{r}_1, \vec{r}_2) \right] \psi(\vec{r}_1, \vec{r}_2, t) = i\hbar \frac{\partial}{\partial t} \psi(\vec{r}_1, \vec{r}_2, t)$$

The two bodies' problem

Suppose that there are no external forces and that the interaction potential $V(\vec{r}_1, \vec{r}_2)$ depends on the relative distances only, i.e.

$$V(\vec{r}_1, \vec{r}_2) \equiv V(|\vec{r}_1 - \vec{r}_2|)$$

Change coordinates :

Center of mass coordinates $\vec{R}$

$$M\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2$$

Relative coordinates $\vec{r}$

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

Momentum of the Center of mass

$$\vec{P} = \vec{p}_1 + \vec{p}_2$$

Relative momentum

$$\vec{p} = \frac{m_2}{M}\vec{p}_1 - \frac{m_1}{M}\vec{p}_2$$

where $M = m_1 + m_2$, $\vec{P} = M\, d\vec{R}/dt$, μ is the reduced mass of the system, $\vec{p} = \mu\vec{v}$ and $\vec{v} = \vec{v}_1 - \vec{v}_2$

The two bodies' problem

Taking into account that $\frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} = \frac{P^2}{2M} + \frac{p^2}{2\mu}$

The 'original' Schrödinger eqn.

$$\left[\frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(|\vec{r}_1 - \vec{r}_2|)\right]\psi(\vec{r}_1, \vec{r}_2, t) = i\hbar\frac{\partial}{\partial t}\psi(\vec{r}_1, \vec{r}_2, t)$$

can be written as (notice that $\psi(\vec{r}_1, \vec{r}_2, t) \implies \Psi(\vec{R}, \vec{r}, t)$)

$$\left[\frac{P^2}{2M} + \frac{p^2}{2\mu} + V(r)\right]\Psi\left(\vec{R}, \vec{r}, t\right) = i\hbar\frac{\partial}{\partial t}\Psi\left(\vec{R}, \vec{r}, t\right)$$

$$P^2 \longrightarrow (-i\hbar\nabla_R)^2 \equiv -\hbar^2\left[\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2}\right]$$

$$p^2 \longrightarrow (-i\hbar\nabla_r)^2 \equiv -\hbar^2\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right]$$

The two bodies' problem

Now, resorting to the method of separation of variables

$$\Psi\left(\vec{R},\vec{r},t\right)=\Phi\left(\vec{R},\vec{r}\right)e^{-iEt/\hbar}=\Phi_R(\vec{R})\Phi_r(\vec{r})e^{-i(E_R+E_r)t/\hbar}$$

we are left with two Schrödinger-like eqns.

$$\underbrace{-\frac{\hbar^2}{2M}\nabla_R^2\Phi_R\left(\vec{R}\right)=E_R\Phi_R\left(\vec{R}\right)}_{\text{Free particle}}$$

Eqn. for a free particle of mass M. Solution is a plane wave with total moment P

$$\underbrace{\left[-\frac{\hbar^2}{2\mu}\nabla_r^2+V(r)\right]\Phi_r(\vec{r})=E_r\Phi_r(\vec{r})}_{\text{Relative motion}}$$

Eqn. for a particle of mass μ moving under the action of the potential, $V(r)$, of interaction between both particles.

Hydrogen's atom

A hydrogen's atom is a system composed of two particles (a proton and an electron) linked by means of a Coulomb's potential.

The proton is a particle with a mass 1836.15 times the electron's mass and has a positive charge of the same magnitude as that of the electron (opposite sign).

The interaction between them is :

$$V(r) = \frac{-e^2}{4\pi\varepsilon_0 r}$$

The subsequent theory can be applied to the more general case of an atom with atomic number Z with all but one of its electrons removed (He^+, Li^{++}, ...), i.e. the so-called hydrogenic atoms.

Now, we would have

$$V(r) = \frac{-Ze^2}{4\pi\varepsilon_0 r}$$

Hydrogen's atom

Now, the radial equation is:

$$-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left(-\frac{Ze^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2\mu r^2}\right)R=ER$$

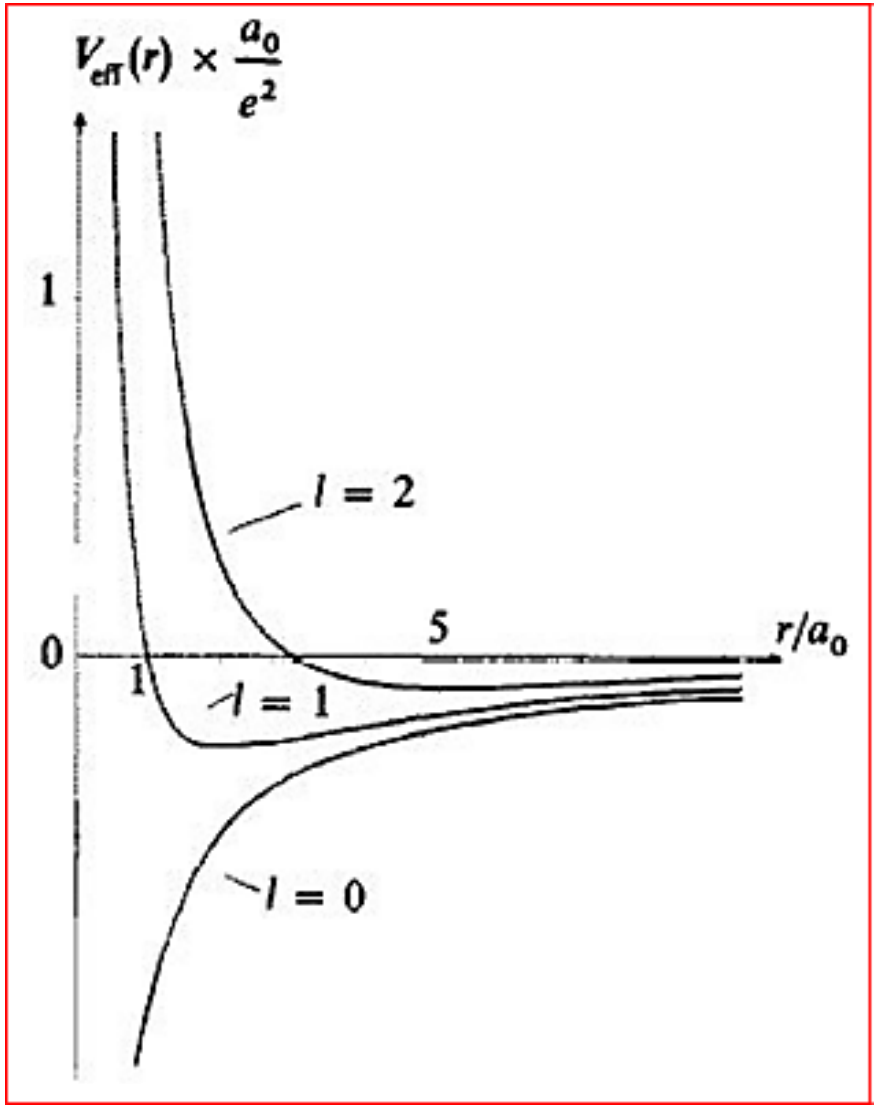
Remember that we started with the expression $V(r)+\lambda/r^2$, but later we changed to $\lambda'=2\mu\lambda/\hbar^2$ and finally we obtained that $\lambda'=l(l+1)$

Where $E<0$, and for a hydrogen-like atom with a nuclear charge Z.

Using the technique of power series and taking into account the appropriate boundary conditions, we finally obtain

$$E_n=-\frac{\mu Z^2 e^4}{(4\pi\varepsilon_0)^2 2\hbar^2 n^2} \qquad \text{con } n=l+1,l+2,\ldots$$

Hydrogen's atom



Shape of the effective potential $V_{\text{eff}}(r)$ for the first values of l in the case where $V(r) = -\frac{e^2}{r}$. When $l = 0$, $V_{\text{eff}}(r)$ is simply equal to $V(r)$. When l takes on the values 1, 2, etc., $V_{\text{eff}}(r)$ is obtained by adding to $V(r)$ the centrifugal potential $\frac{l(l+1)\hbar^2}{2\mu r^2}$, which approaches $+\infty$ when r approaches zero.

Remember that $\mathcal{X}(r) = rR(r)$

Hydrogen's atom

$$-\frac{\hbar^2}{2\mu}\frac{d^2\mathcal{X}}{dr^2}+\left(-\frac{Ze^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2\mu r^2}\right)\mathcal{X}=E\mathcal{X}$$

where $\mathcal{X}(r)=rR(r)$ and we define a new variable $\rho=\left(-8\mu E/\hbar^2\right)^{1/2}r$

$$\frac{d^2\mathcal{X}}{d\rho^2}-l(l+1)\frac{\mathcal{X}}{\rho^2}+\left(\frac{\beta}{\rho}-\frac{1}{4}\right)\mathcal{X}=0$$

where

$$\beta=\left(-\frac{\mu}{2E}\right)^{1/2}\frac{Ze^2}{4\pi\varepsilon_0\hbar}$$

First, we consider the asymptotic behavior of the solution. In this case the eqn. becomes

$$\frac{d^2\mathcal{X}}{d\rho^2}-\frac{1}{4}\mathcal{X}=0$$

Hydrogen's atom

The solution's asymptotic behavior would be

$$\mathcal{X}(\rho) \propto e^{-\rho/2}$$

we have not included the solution with positive exponent.

We search for a solution ...

$$\mathcal{X}(\rho) = F(\rho)e^{-\rho/2}$$

and write

$$F(\rho) = \sum_{p=1}^{\infty} a_p \rho^p$$

Notice that $a_0 = 0$, because $\mathcal{X}(\rho \to 0) = 0$. Otherwise, there would be a divergence at $r = 0$.

Hydrogen's atom

Substituting we get

$$-l(l+1)a_1\rho^{-1} + \sum_{p=1}^{\infty}\left[(p+1)pa_{p+1} - pa_p - l(l+1)a_{p+1} + \beta a_p\right]\rho^{p-1} = 0$$

The coefficient of each power must vanish, so

$a_1 = 0$ *unless* $l = 0$ (*i.e. if* $l \neq 0 \rightarrow a_1 = 0$)

and

$$\frac{a_{p+1}}{a_p} = \frac{p-\beta}{p(p+1) - l(l+1)}$$

This ratio tends to $^1/_p$ as $p \rightarrow \infty$

Hydrogen's atom

According to the previous recursion relation, the denominator becomes zero when $p = l$

This would imply that $a_{l+1} \to \infty$, UNLESS $a_l = 0$

But if $a_l = 0$, then $a_{l-1} = a_{l-2} = \cdots = 0$

That is, $a_p = 0 \quad \forall\, p \leq l \quad \Longrightarrow$ **The series begins with the term a_{l+1}**

In summary

If $l = 0$ *then* $a_1, a_2, a_3, a_4, a_5 \ldots \neq 0$

If $l = 1$ *then* $a_1 = 0$, and $a_2, a_3, a_4, a_5 \ldots \neq 0$

If $l = 2$ *then* $a_1 = a_2 = 0$, and $a_3, a_4, a_5 \ldots \neq 0$

If $l = 3$ *then* $a_1 = a_2 = a_3 = 0$ and $a_4, a_5 \ldots \neq 0$

If $l = l_f$ *then* $a_1 = a_2 = a_3 = \cdots = a_{l_f} = 0$ and $a_{l_f+1}, a_{l_f+2}, a_{l_f+3} \ldots \neq 0$

Then, we can write $F(\rho) = \sum_{p=l_{f+1}}^{\infty} a_p \rho^p$

Hydrogen's atom

On the other hand, the series behaves (when $p \to \infty$) like e^{ρ}.

Then, $\mathcal{X}(\rho) = F(\rho)e^{-\rho/2}$ would diverge as $e^{\rho/2}$

SOLUTION: Terminate the series!

For the term $p = n$ $\left(n \geq (l+1)\right)$, we must have $\beta = n \geq (l+1)$

So we finally have

$$F(\rho) \equiv F_n(\rho) = \sum_{p=l+1}^{n} a_p \rho^p$$

Hydrogen's atom

Which can be written as

$$F(\rho) \equiv F_n(\rho) = \rho^{l+1}\left[a_{l+1} + a_{l+2}\rho + a_{l+3}\rho^2 + \cdots + a_n\rho^{n-(l+1)}\right] \equiv \rho^{l+1}L_{n-(l+1)}(\rho)$$

with

$$L_k(\rho) = b_0 + b_1\rho + b_2\rho^2 + \cdots + b_k\rho^k \qquad b_0 \neq 0$$

The $L_k(\rho)$ are the Laguerre polynomials

$$L_0(\rho) = 1 \qquad L_1(\rho) = 1 - \rho \qquad L_2(\rho) = \frac{1}{2}\left(2 - 4\rho + \rho^2\right) \qquad L_3(\rho) = \frac{1}{6}\left(6 - 18\rho + 9\rho^2 - \rho^3\right)$$

Hydrogen's atom

Noticed that we imposed $\beta = n$.

We obtain energy quantization

But $\beta = \left(-\frac{\mu}{2E}\right)^{1/2} \frac{Ze^2}{4\pi\varepsilon_0\hbar}$

Finally, the radial function $R(r)$ is obtained

But notice that it depends on two indexes (quantum numbers) n and l

So we will write it as $R_{n,l}(r)$

Moreover, for a fixed n-value, there are several l -values, namely $(n-(l+1) \geq 0)$

$$l = 0,1,2,\ldots,n-1$$

Notice:

$$\int_0^\infty r^2 R^*_{n,l}(r) R_{n',l}(r)\, dr = \delta_{n,n'} \qquad \text{(same } l\text{)}$$

and

$$\int_0^\infty r^2 \left|R_{n,l}(r)\right|^2 dr = 1$$

For different l's, the above integral can give any result.

Hydrogen's atom

In other words, we have found that

$$|m| = 0,1,2,3, \ldots$$

$$l = |m|, |m| + 1, |m| + 2, |m| + 3, \ldots$$

$$n = l + 1, l + 2, l + 3, \ldots$$

which is equivalent to

$$n = 1, 2, 3, \ldots$$

$$l = 0, 1, 2, \ldots, n - 1$$

$$m = -l, -l + 1, -l + 2, \ldots, 0, \ldots, l - 1, l$$

Hydrogen's atom

For hydrogen's atom, the eigenfunctions (solutions of the time-independent Schrödinger's equation) are

$$\psi_{n,l,m}(r,\theta,\phi) = R_{n,l}(r)Y_l^m(\theta,\phi)$$

and the respective eigenvalues

$$E_n = -\frac{\mu e^4 Z^2}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}$$

The three quantum numbers are:

$n = 1, 2, 3, \ldots$ $\quad$ $l < n$ $\quad$ $-l \leq m \leq l$

Finally,

$$\psi_{n,l,m}(r,\theta,\phi,t) = R_{n,l}(r)Y_l^m(\theta,\phi)e^{-iE_n t/\hbar}$$

Hydrogen's atom. Eigenfunctions

If we write

$$\psi(\mathrm{r},\theta,\phi) = R_{n,l}(r)\Theta_{l,m}(\theta)\Phi_m(\phi)$$

Recall that

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$$

$$\Theta_{l,m}(\theta) \propto sin^{|m|}\theta \underbrace{(polynomial\ in\ \cos\theta)}_{degree\ \ l-|m|}$$

$$R_{n,l}(r) \propto e^{-Constant\frac{r}{n}} r^l \underbrace{(polynomial\ in\ r)}_{degree\ \ n-(l+1)}$$

$$R_{n,l}(r) \propto e^{-Zr/na_0} \left(\frac{Zr}{a_0}\right)^l G_{n,l}\left(\frac{Zr}{a_0}\right)$$

Hydrogen's atom. Eigenfunctions

Quantum Numbers n	l	m_l	Eigenfunctions
1	0	0	$\psi_{100} = \frac{1}{\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}$
2	0	0	$\psi_{200} = \frac{1}{4\sqrt{2\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2-\frac{Zr}{a_0}\right) e^{-Zr/2a_0}$
2	1	0	$\psi_{210} = \frac{1}{4\sqrt{2\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\frac{Zr}{a_0} e^{-Zr/2a_0}\cos\theta$
2	1	± 1	$\psi_{21\pm1} = \frac{1}{8\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\frac{Zr}{a_0} e^{-Zr/2a_0}\sin\theta\, e^{\pm i\varphi}$

Hydrogen's atom. Eigenfunctions

3	0	0	$\psi_{300} = \frac{1}{81\sqrt{3\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(27 - 18\frac{Zr}{a_0} + 2\frac{Z^2r^2}{a_0^2}\right)e^{-Zr/3a_0}$
3	1	0	$\psi_{310} = \frac{\sqrt{2}}{81\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(6 - \frac{Zr}{a_0}\right)\frac{Zr}{a_0}e^{-Zr/3a_0}\cos\theta$
3	1	± 1	$\psi_{31\pm 1} = \frac{1}{81\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(6 - \frac{Zr}{a_0}\right)\frac{Zr}{a_0}e^{-Zr/3a_0}\sin\theta\, e^{\pm i\varphi}$
3	2	0	$\psi_{320} = \frac{1}{81\sqrt{6\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\frac{Z^2r^2}{a_0^2}e^{-Zr/3a_0}(3\cos^2\theta - 1)$
3	2	± 1	$\psi_{32\pm 1} = \frac{1}{81\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\frac{Z^2r^2}{a_0^2}e^{-Zr/3a_0}\sin\theta\cos\theta\, e^{\pm i\varphi}$
3	2	± 2	$\psi_{32\pm 2} = \frac{1}{162\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\frac{Z^2r^2}{a_0^2}e^{-Zr/3a_0}\sin^2\theta\, e^{\pm 2i\varphi}$

Hydrogen's atom. Eigenvalues

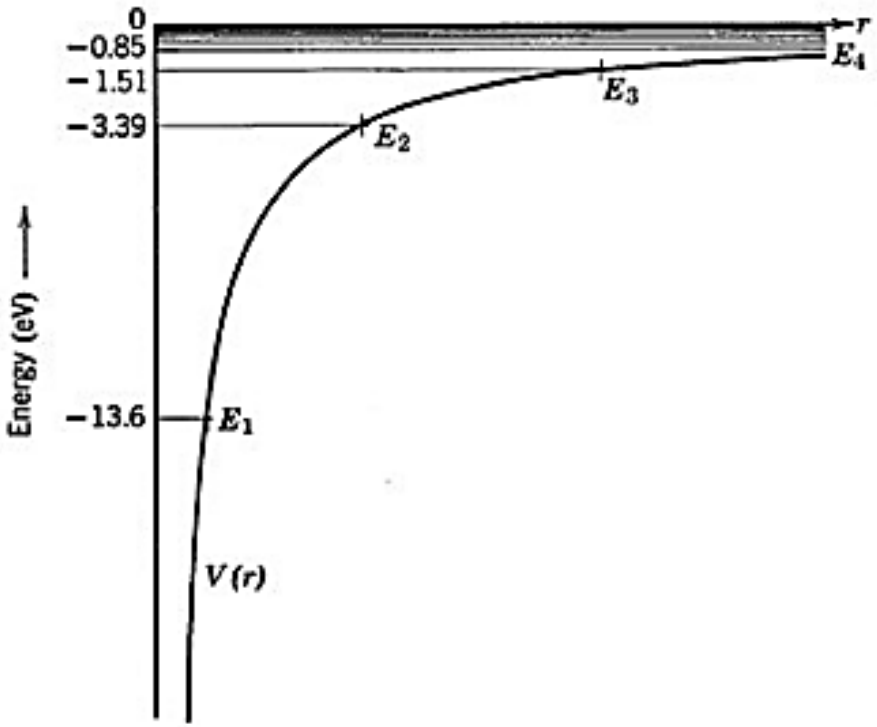
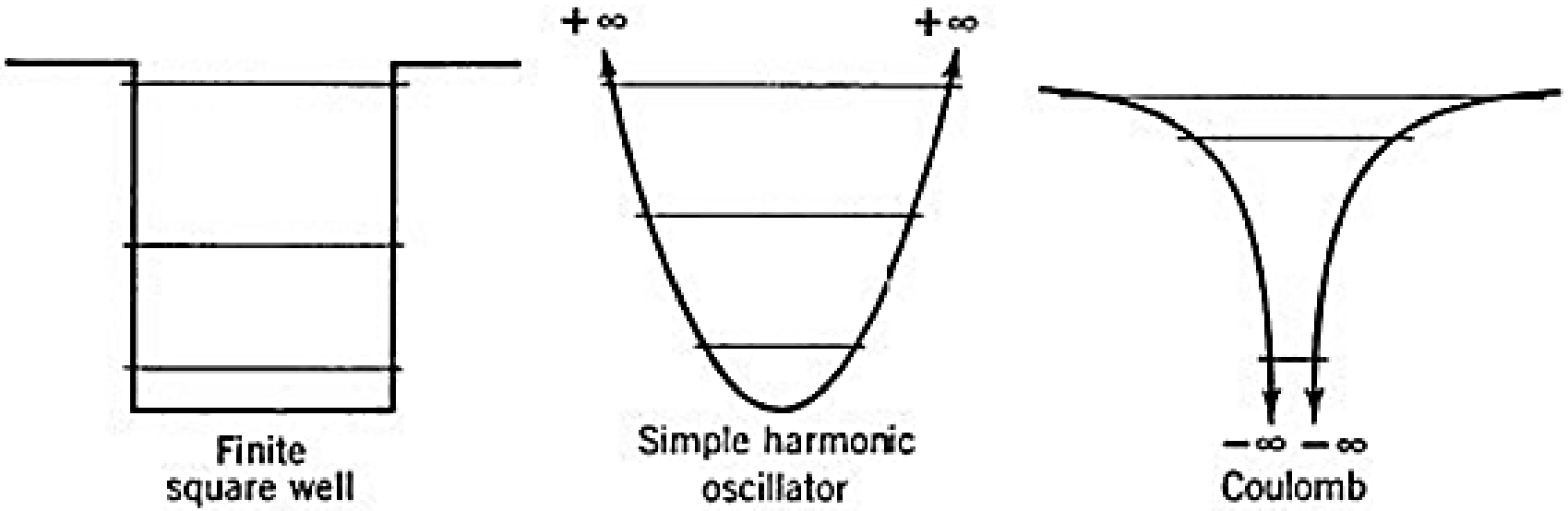


Figure 7-3 The Coulomb potential $V(r)$ and its eigenvalues E_n. For large values of n the eigenvalues become very closely spaced in energy since E_n approaches zero as n approaches infinity. Note that the intersection of $V(r)$ and E_n, which defines the location of one end of the classically allowed region, moves out as n increases. Not shown in this figure is the continuum of eigenvalues at positive energies corresponding to unbound states.

Hydrogen's atom. Eigenvalues



A comparison between the allowed energies of several binding potentials. The three-dimensional Coulomb potential is shown in a cross-sectional view along a diameter; the other potentials are one-dimensional.

Hydrogen's atom. Degeneracy

The eigenvalues (energies) depend on the quantum number n only, BUT the eigenfunctions depend on the three quantum numbers n, l, m.

For a given n (which means a fixed energy value) there are several different eigenfunctions with the same eigenvalue. In fact, we have all the eigenfunctions

$$R_{n,l}(r)Y_l^m(\theta,\phi)$$

which amount to a total $\sum_{l=0}^{n-1}(2l+1) = n^2$ different functions which are orthogonal.

$$\int_0^{\infty}\int_0^{\pi}\int_0^{2\pi} r^2 \sin\theta \; dr\, d\theta\, d\phi \left[R_{n,l}(r)Y_l^m(\theta,\phi)\right]^* R_{n,q}(r)Y_q^k(\theta,\phi) = \; \delta_{l,q}\,\delta_{m,k}$$

$R_{n,l}(r)$ and $R_{n,q}(r)$ are the solutions of two different differential equations!!!

The addition of the n terms of an arithmetic sequence gives:

$$S = \frac{n(a_1 + a_n)}{2} = \frac{n(1 + 2n - 2 + 1)}{2} = n^2$$

Hydrogen's atom. Degeneracy

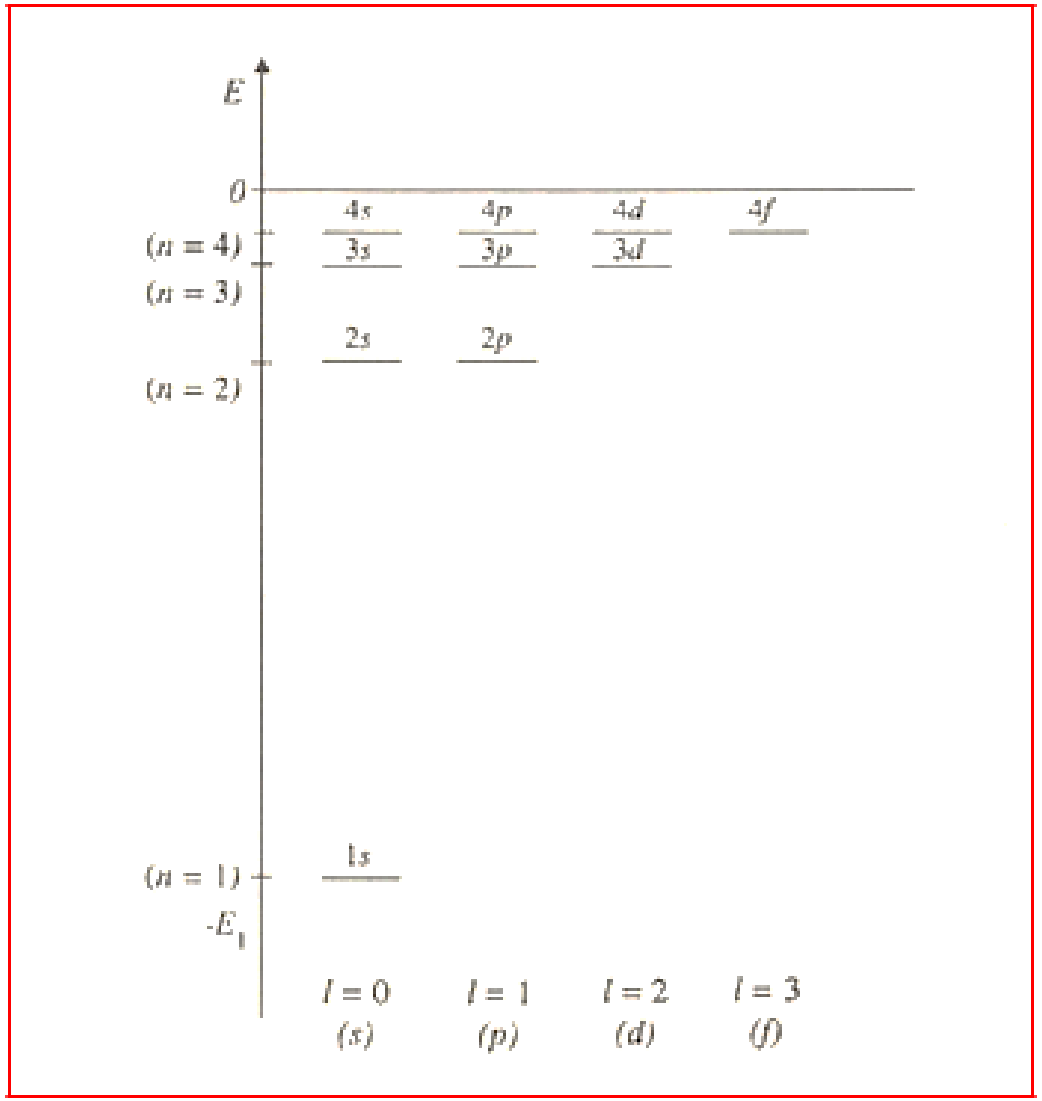
So, we have that two or more completely different eigenfunctions correspond exactly to the same eigenvalue E_n.

But the eigenfunctions describe the behaviour of the (electron) atom. So, it may happen that states with completely **different** behaviour have the **same** total energy.

This phenomenon is called degeneracy

The eigenfunctions are said to be degenerate

Hydrogen's atom



- for fixed *n* there are n possible *l* -values
- for fixed *l* there are $2l + 1$ possible *m*-values
- for fixed *n* there are n^2 possible (l, m)-values
- for fixed *n* and *l* , the different *m* values have the same energy (essential degeneracy)
- for fixed *n* and variable *l* , the eigenfunctions have same energy (accidental degeneracy)

Hydrogen's atom. Probability density

According to the probabilistic interpretation of the wave function

$$\left|\psi_{n,l,m}(r,\theta,\phi)e^{-iE_n t/\hbar}\right|^2 r^2 \sin\theta \; d\theta \; d\phi \; dr$$

gives the probability of finding the electron inside the differential volume element centered at the point (r,θ,ϕ)

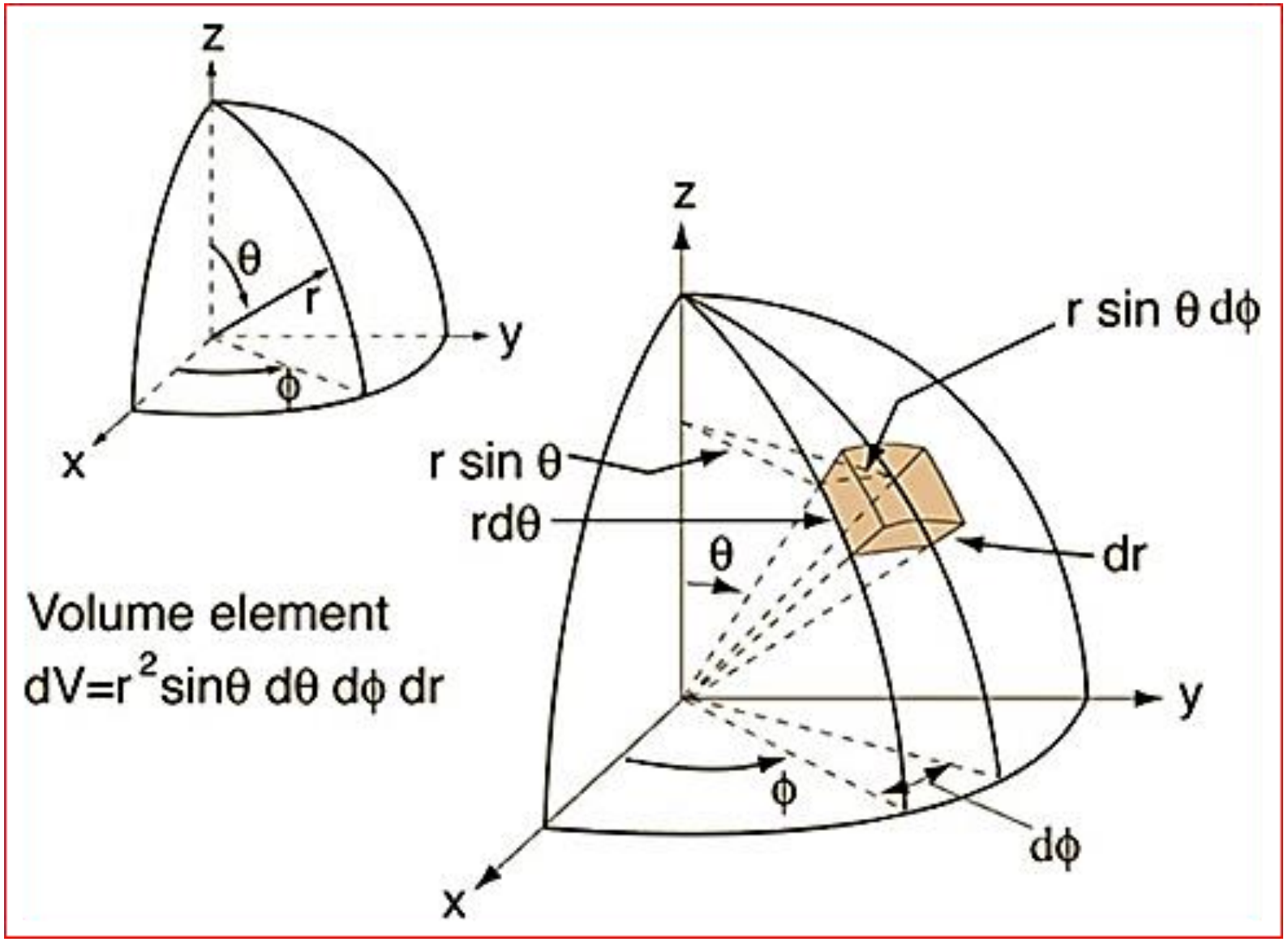
How does the probability change as a function of the distance to the origin? (radial dependence).

Integrate the previous equation with respect to the angular dependence:

$$P_{n,l}(r)dr = \int d\Omega \left|\psi_{n,l,m}(r,\theta,\phi)\right|^2 r^2 \, dr = \underbrace{\int d\Omega \, |Y_l^m(\theta,\phi)|^2}_{=1} \left|R_{n,l}(r)\right|^2 r^2 \, dr$$

Radial probability density. It is the probability of finding the (n, l)-electron with the radial coordinate inside the range $[r, r + dr]$

Volume element in spherical coordinates

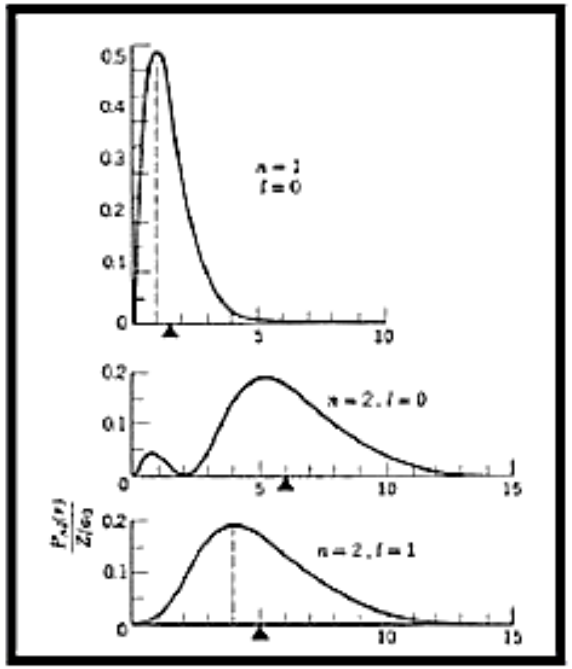


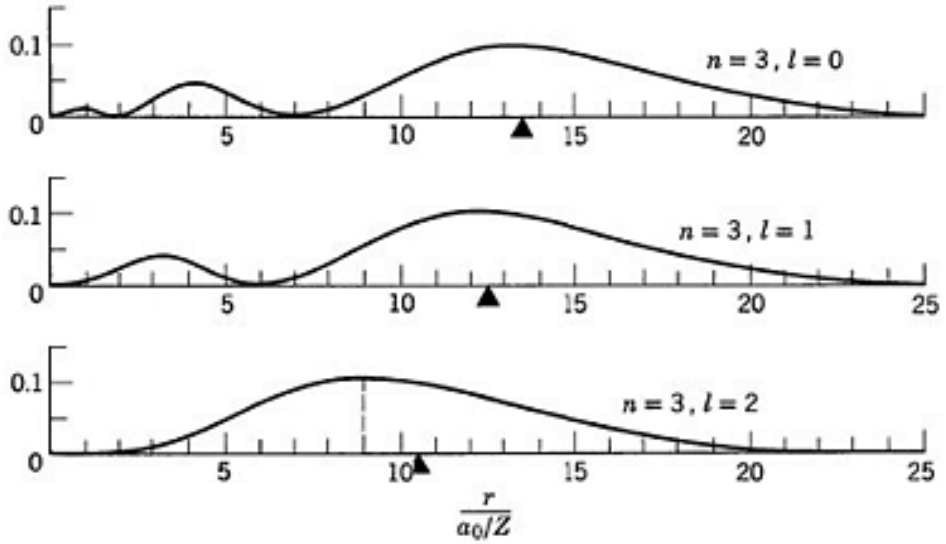
Hydrogen's atom. Eigenvalues

Radial probability density $P_{n,l}(r) \equiv r^2 R_{n,l}(r) R^*_{n,l}(r) \equiv r^2 \left| R_{n,l}(r) \right|^2$

Remember that $R_{n,l}(r) \propto \left(r^l\right) \cdot (polynomial\ degree\ [n-(l+1)])$

Therefore, it crosses the axis $[n-(l+1)]$ times

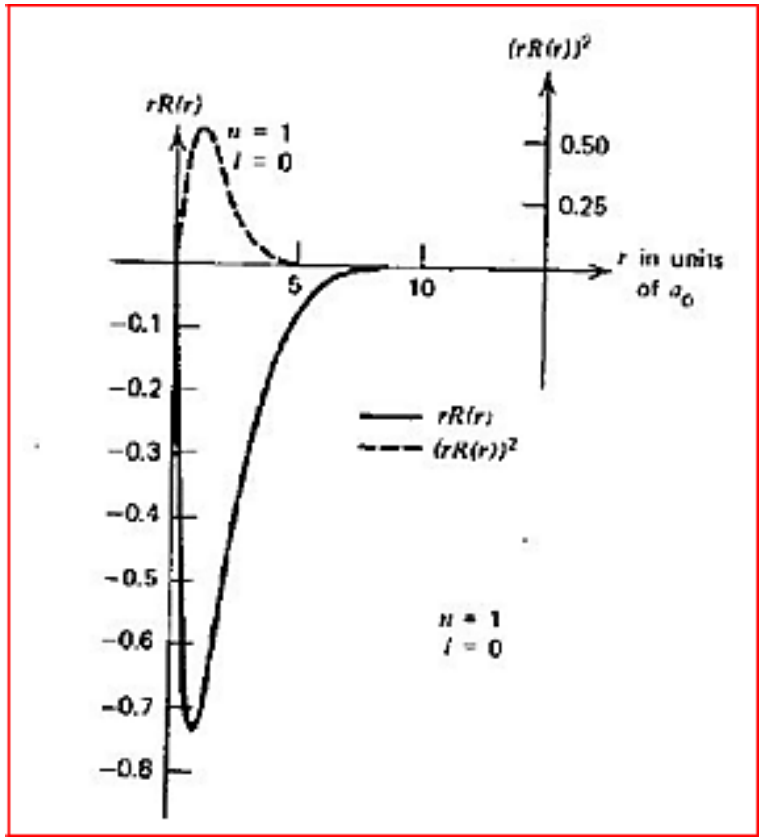


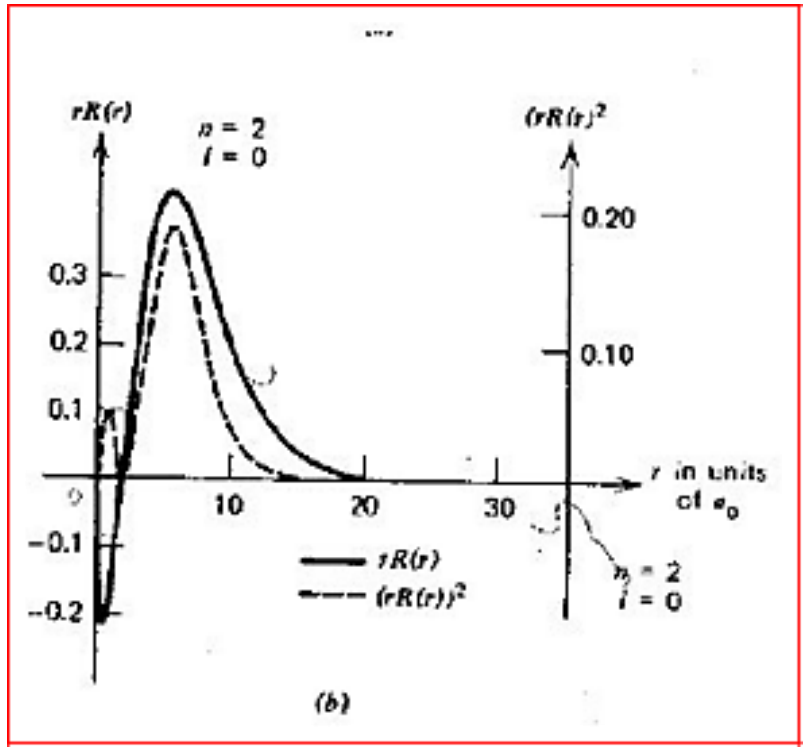


---- Bohr's radius

▲ mean value

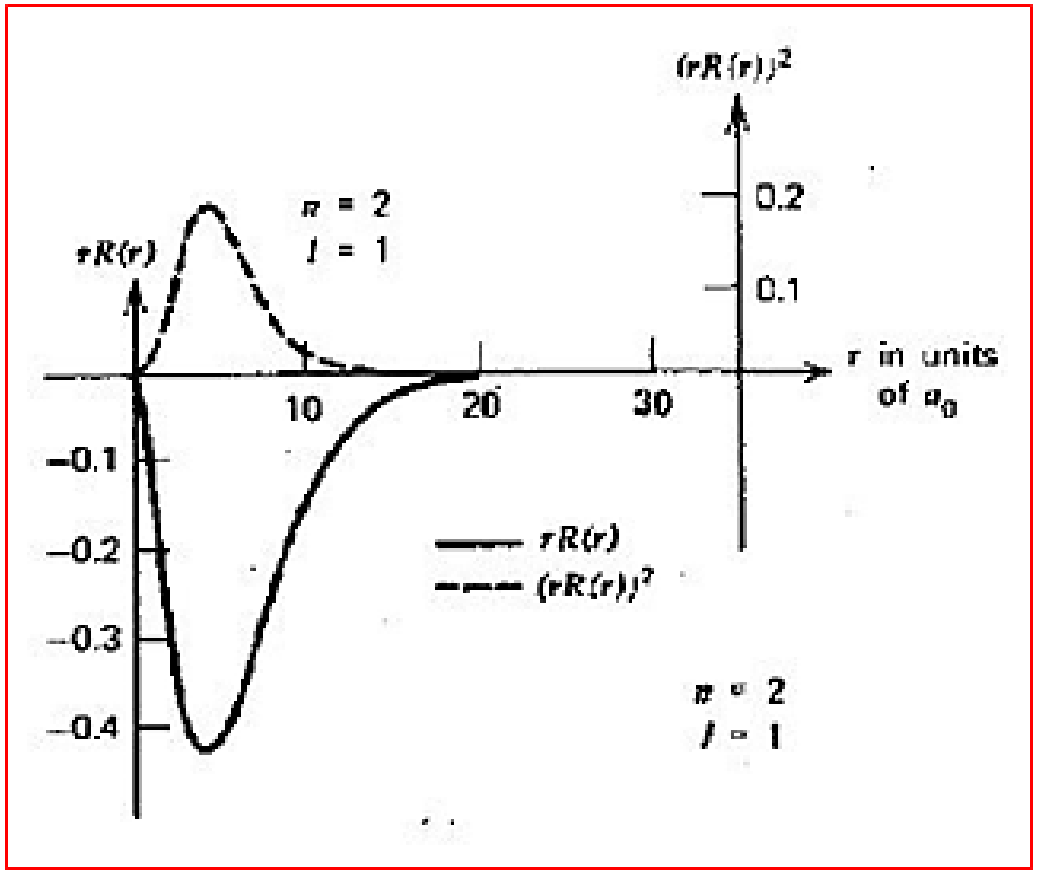
Hydrogen's atom

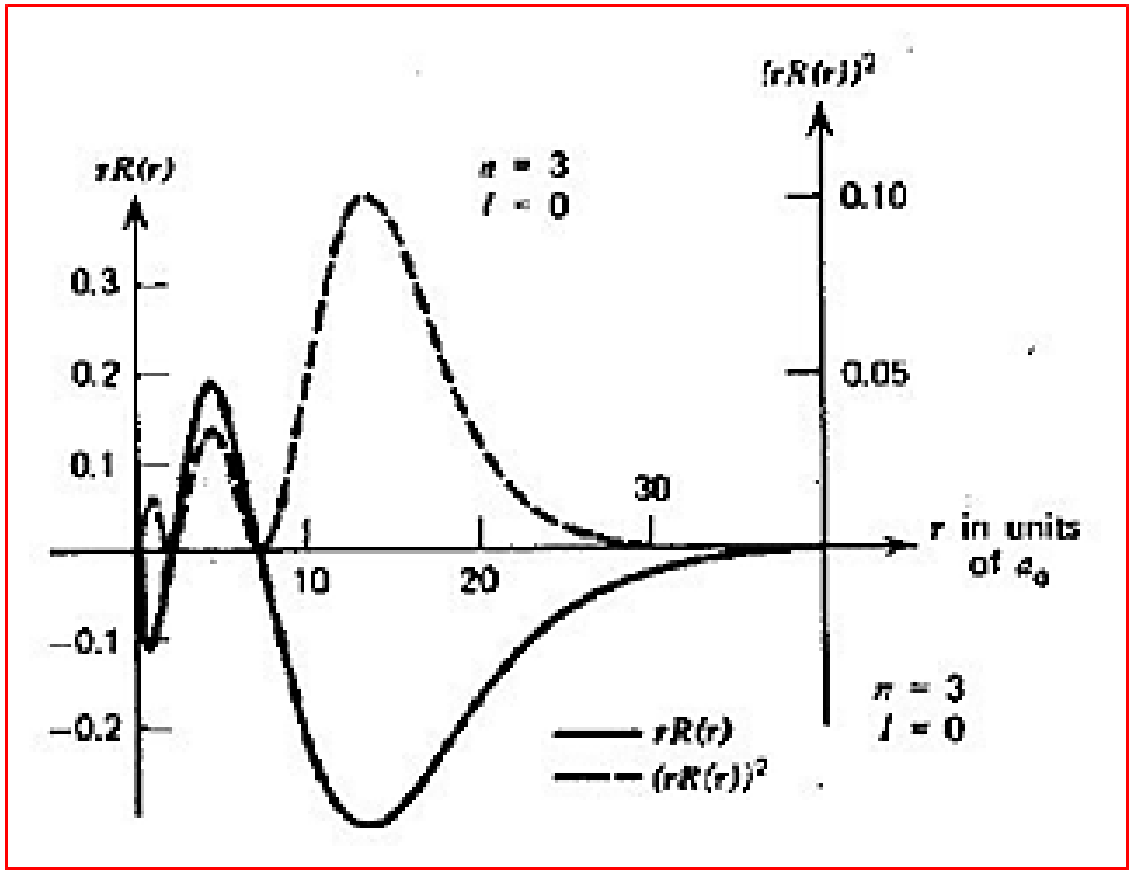




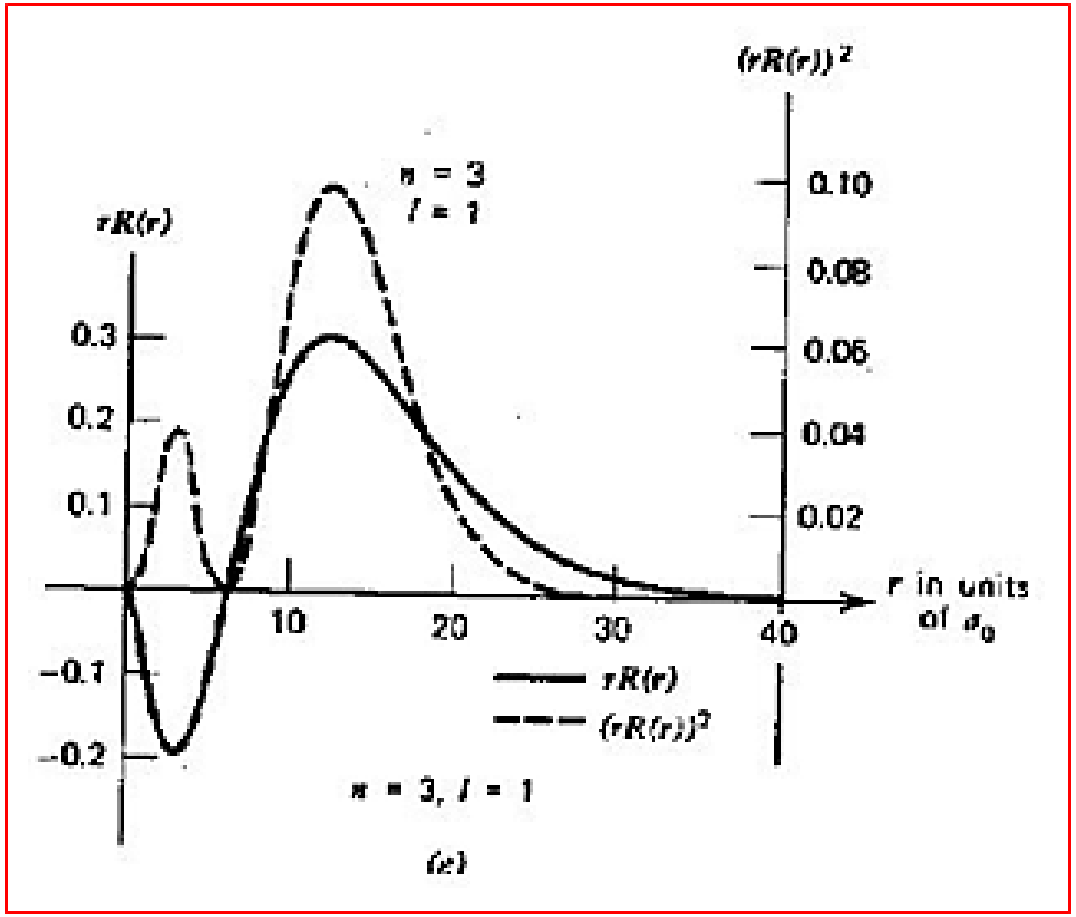
The radial wave functions $u(r) = rR(r)$ and the radial probability density function $u^2(r)$ for values of $n = 1, 2, 3, 4$ and the values of l possible. The left abscissa measures $u(r)$ and the right abscissa measures $u^2(r)$. The wave functions are given by the solid lines, and the probability distributions by the dashed lines. The ordinate measures r in units of a_0.

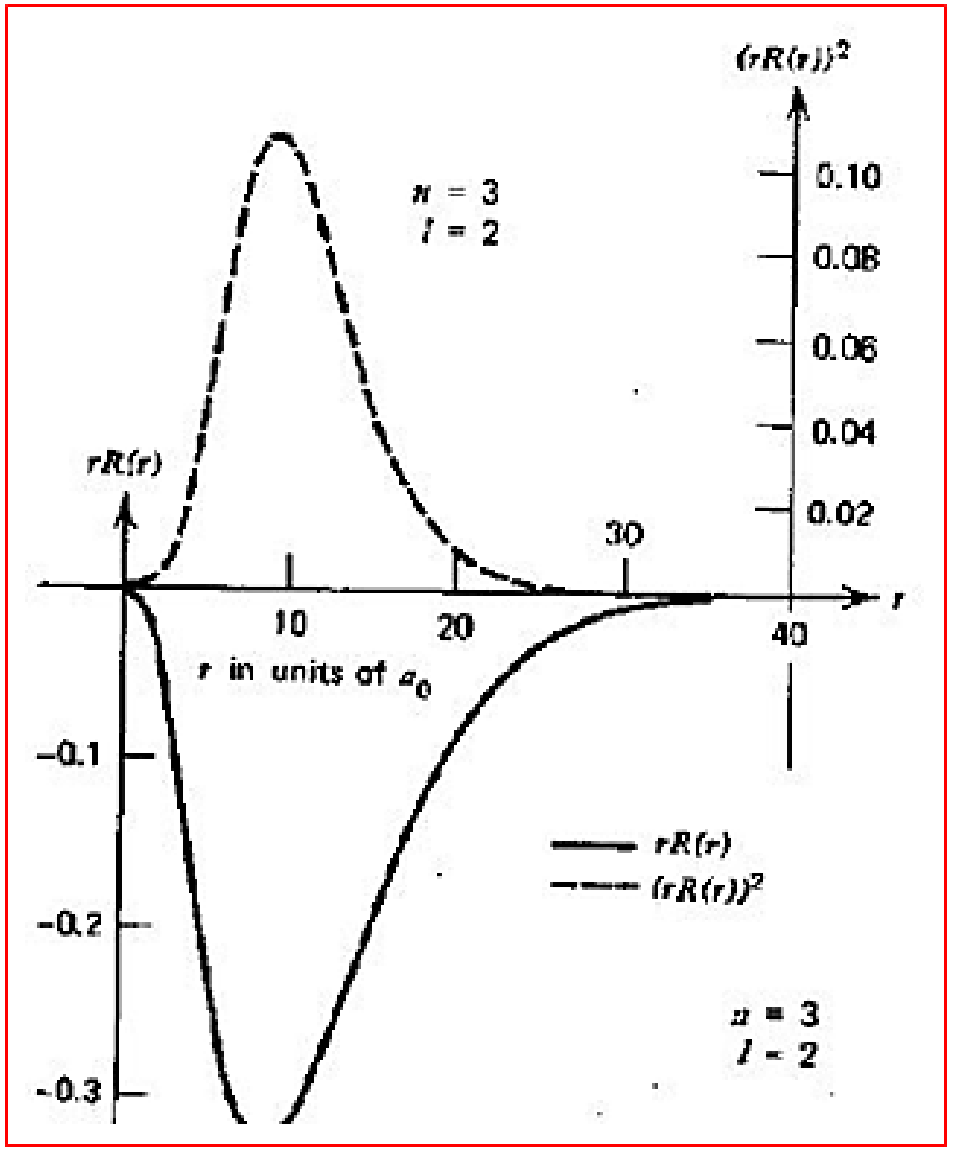
Hydrogen's atom



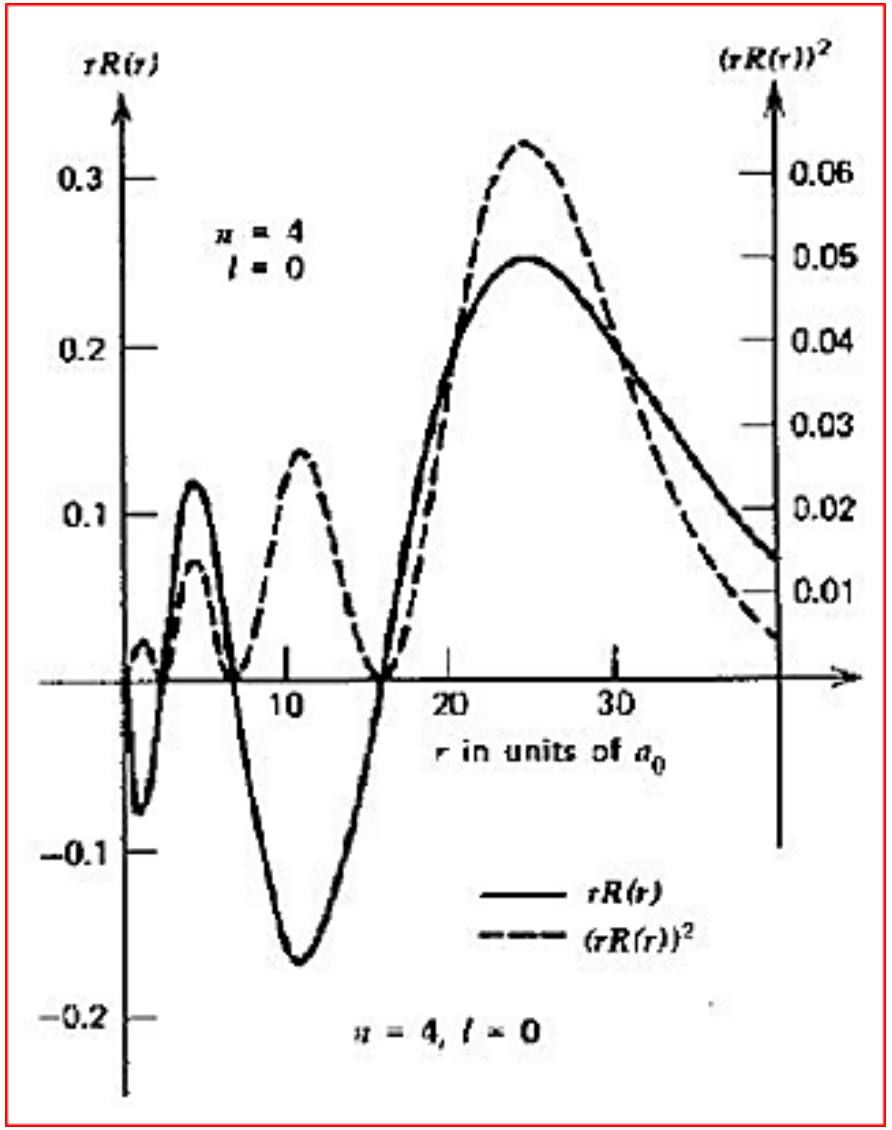


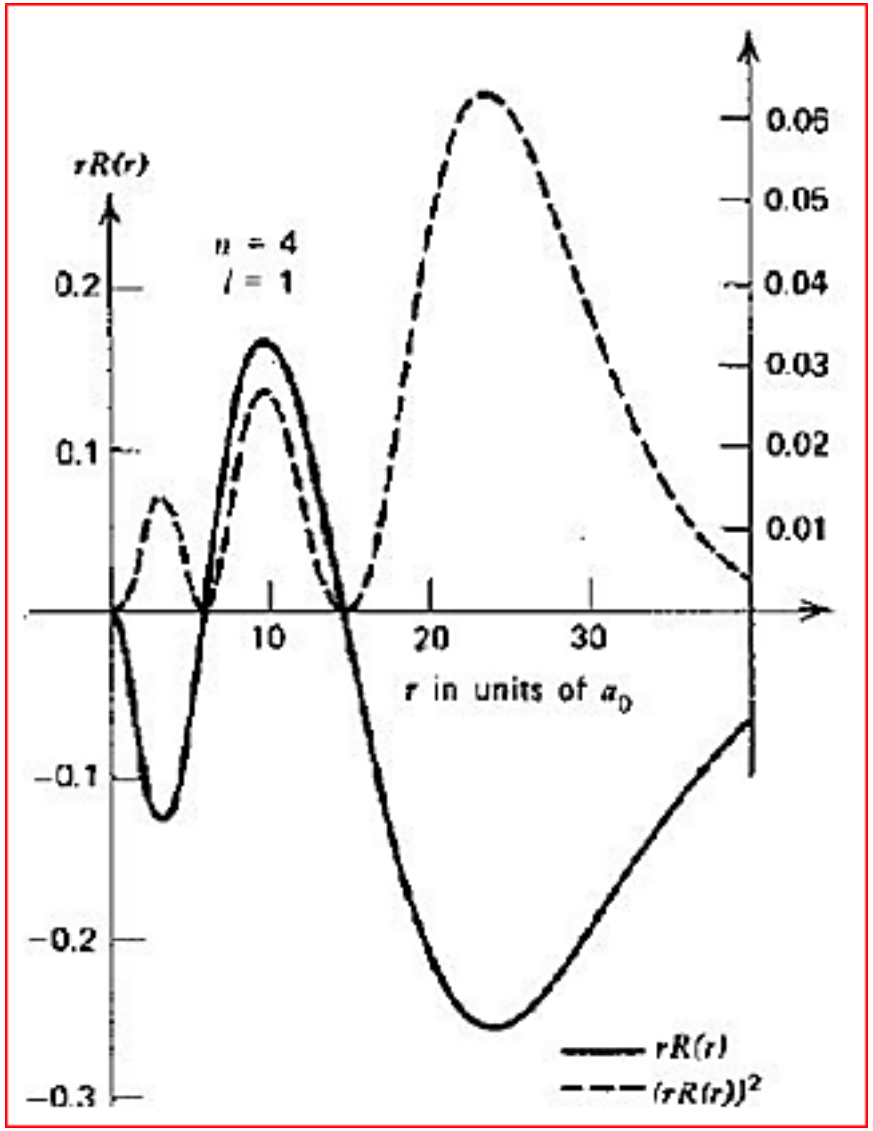
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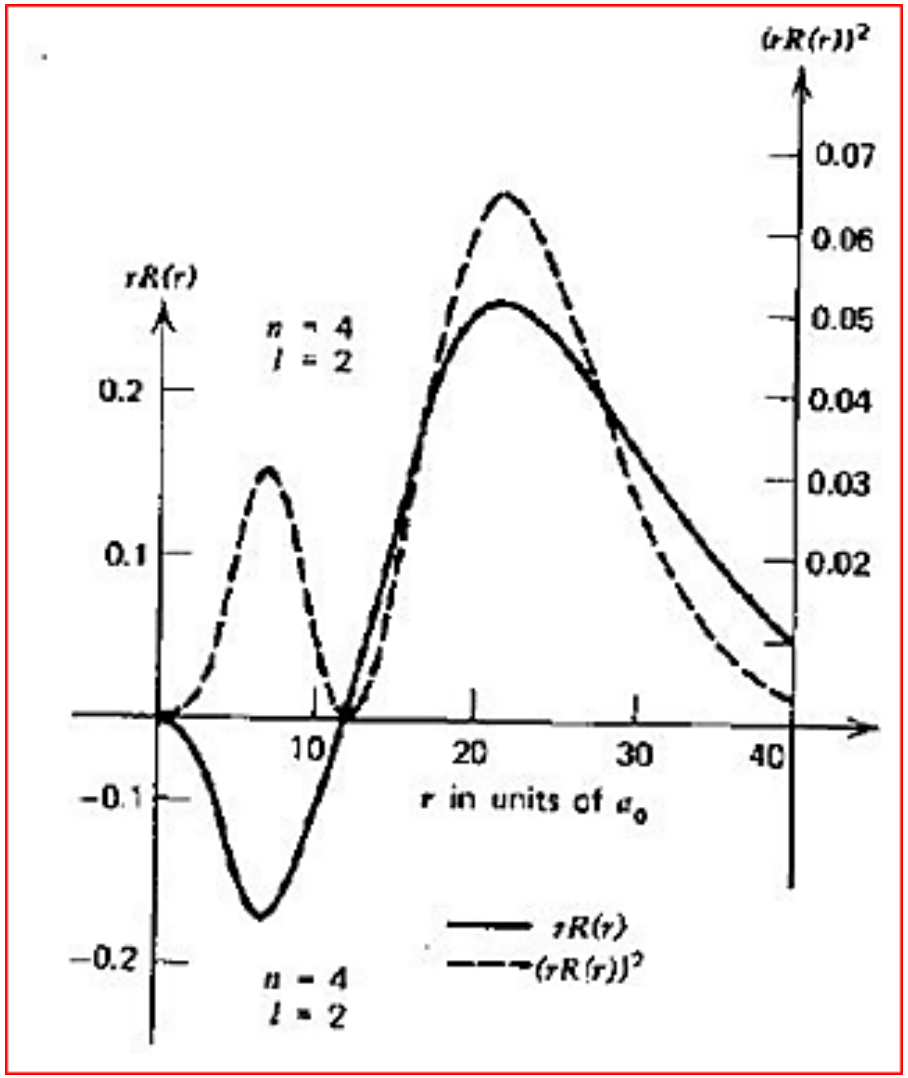


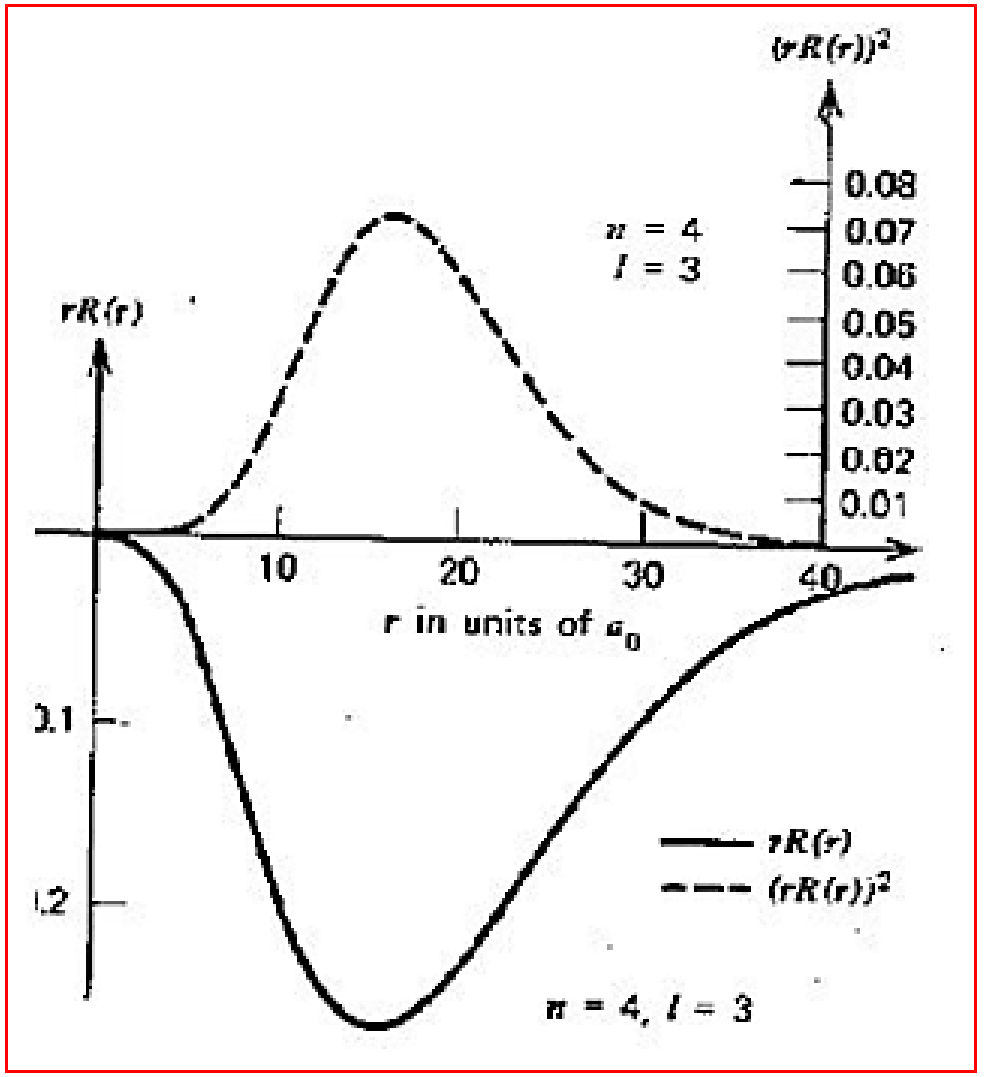
Hydrogen's atom





Hydrogen's atom





Hydrogen's atom. Probability density

The radial probability densities have significant values within some range of the radial coordinate.

The location of the range depends, mainly, on the quantum number *n* (there is also a weak dependence on the *l*-value)

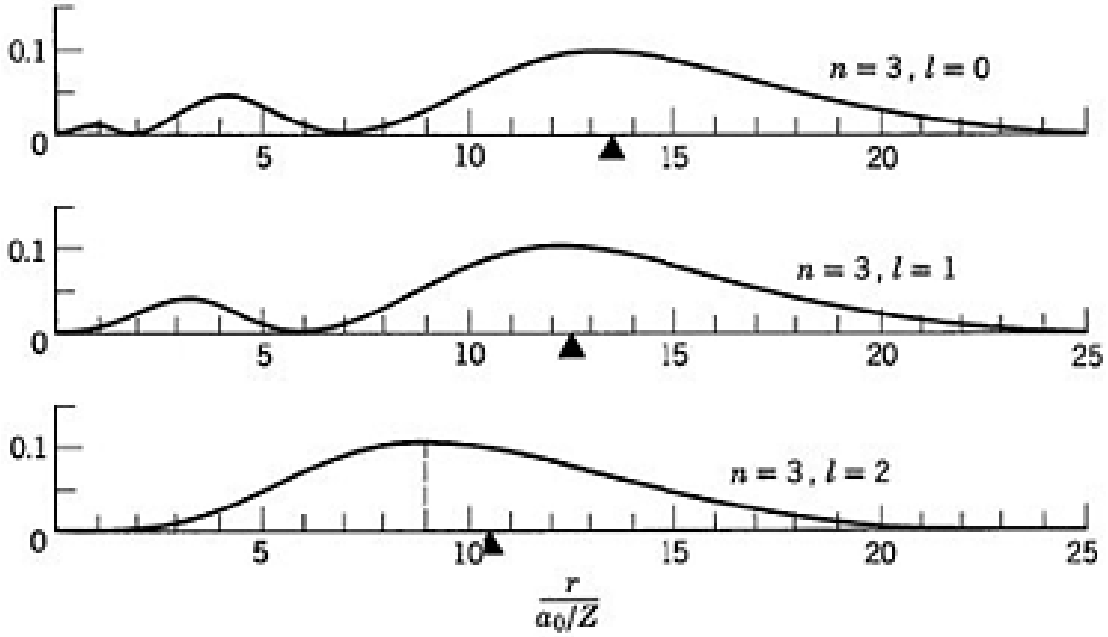
This is clearly evinced by the mean value of the radial coordinate:

$$\bar{r}_{n,l} = \int d^3r \;\; \psi^*_{n,l,m}(r,\theta,\phi)\, r\, \psi_{n,l,m}(r,\theta,\phi) = \int_0^\infty r\, P_{n,l}(r)\, dr = n^2 \frac{a_0}{Z}\left[1 + \frac{1}{2}\left(1 - \frac{l(l+1)}{n^2}\right)\right]$$

Compare with Bohr's radius

$$r_{Bohr} = \frac{n^2 a_0}{Z}$$

Hydrogen's atom. Probability density

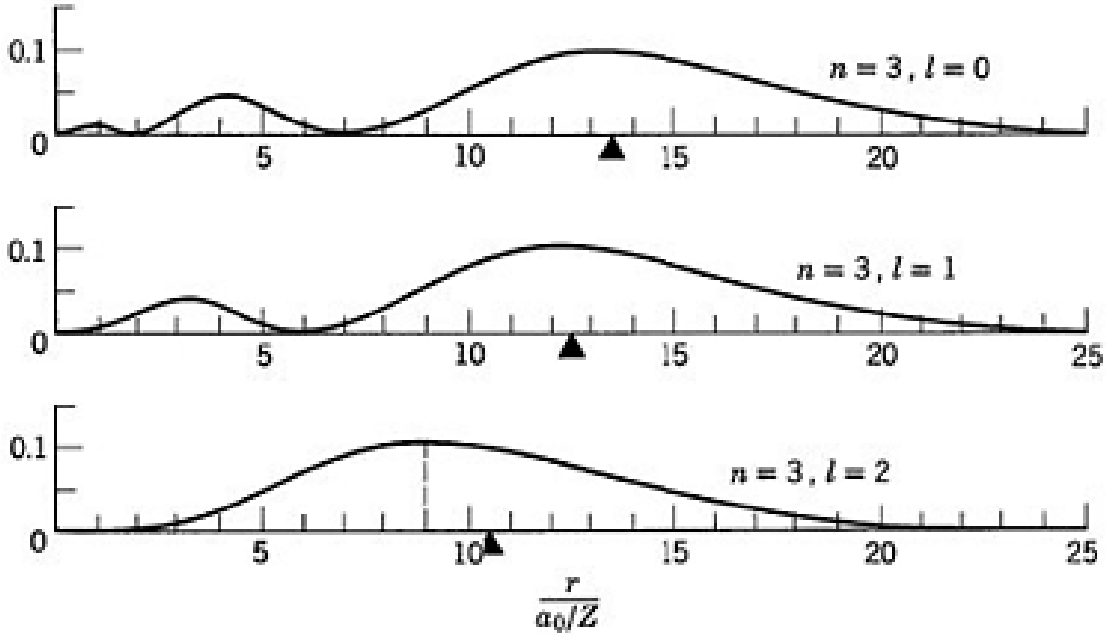


Dashed line: Bohr's model Triangle (▲): Mean value

When $l = 0$ then $\bar{r}_{n,0} = \frac{3}{2Z} n^2 a_0$ and $(\Delta r)_{n,0} = \sqrt{n^2 + 2}\,\frac{n a_0}{2Z}$

When $l = n - 1$ then $\bar{r}_{n,n-1} = (2n + 1)\,\frac{1}{2Z} n a_0$ and $(\Delta r)_{n,n-1} = \sqrt{2n + 1}\,\frac{n a_0}{2Z}$

Hydrogen's atom. Probability density



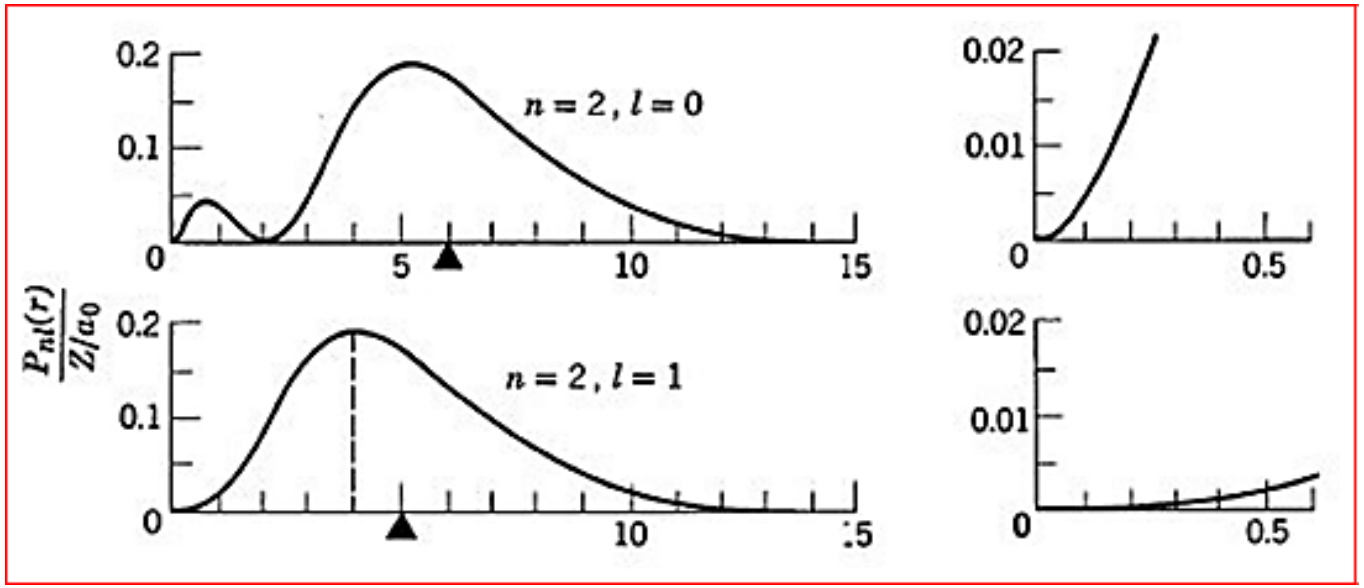
The radial probability density depends on the *l* value too.

When $l = n - 1$ the radial probability density $P_{n,n-1}(r)$ has just one maximum and moreover, it is located at

$$(r_{max})_{n,n-1} = n^2 a_0$$

Hydrogen's atom. Probability density

Small *r* behaviour



We know that $\psi(r, \theta, \phi) \propto r^l$ (when $r \to 0$) and therefore $\psi^*\psi \propto r^{2l}$

$\psi^*\psi$ in a small volume around $r = 0$ takes significant values only when $l = 0$

There is some probability that the electron be near the nucleus if $l = 0$

Hydrogen's atom. Summary

$$\psi_{n,l,m}(r,\theta,\phi) = R_{n,l}(r)Y_l^m(\theta,\phi) \qquad n = 1,2,\ldots \qquad l < n \qquad -l \le m \le l$$

$$E_n = -\frac{\mu Z^2 e^4}{(4\pi\varepsilon_0)^2 2\hbar^2 n^2}$$

$$R_{n,l}(r) = \frac{2}{n^2}\left(\frac{Z}{a_0}\right)^{3/2}\left[\frac{(n-l-1)!}{(n+l)!}\right]^{1/2} e^{-\frac{Zr}{na_0}}\left(\frac{Zr}{na_0}\right)^l L_{n-l-1}\left(\frac{Zr}{na_0}\right)$$

where $L_{n-l-1}(Zr/na_0)$ is Laguerre's polynomial and a_0 is Bohr's radius.

$$Y_l^m(\theta,\phi) = \left[\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}\right]^{1/2} P_l^{|m|}(\cos\theta)e^{im\phi}$$

where $P_l^{|m|}(\cos\theta)$ are Legendre's polynomials